Physics — Class 12 (PCTB, English Medium)

Solved Exercises | Punjab Curriculum and Textbook Board | Chapters 12 to 21 | ALL chapters exercises COMPLETE

FULL-EXERCISE NOTE: Chapters 12 (book pp. 28-30), 13 (book pp. 53-55), 14 (book pp. 79-80), 15 (book pp. 107-110), 16 (book pp. 132-134), 17 (book pp. 152-153), 18 (book pp. 172-174), 19 (book pp. 199-201), 20 (book pp. 217-218) and 21 (book pp. 258-259) are now complete: every QUESTION, MCQ and PROBLEM in this book. Problems 12.2, 15.4, 17.3/17.4/17.6, 19.3, 21.5 and 21.6: see the notes inside those items about the book's printed answers.

Chapter 12: Electrostatics

Chapter opener - Coulomb's law (kept from the earlier build)

State Coulomb's law and write its mathematical form.

Coulomb's law states that the force between two point charges is directly proportional to the product of their magnitudes and inversely proportional to the square of the distance between them.

Mathematically, F is proportional to q1 q2 / r squared.

In free space, F = (1 / 4 pi epsilon-zero) x (q1 q2 / r squared).

The constant 1 / 4 pi epsilon-zero equals 9 x 10^9 N m^2 C^-2.

Exercise - QUESTIONS 12.1 to 12.9 (book p. 28, verbatim)

12.1 The potential is constant throughout a given region of space. Is the electrical field zero or non-zero in this region? Explain.

The electric field is zero in this region.

The field is the negative rate of change of potential with distance, E = - (change in V) / (change in r).

If V is constant, its change is zero, so E = 0.

There is no field where the potential does not vary from point to point.

12.2 Suppose that you follow an electric field line due to a positive point charge. Do electric field and the potential increase or decrease?

Both decrease as you move away from the positive charge.

The field of a point charge is E = kQ / r squared, which falls as r grows.

The potential is V = kQ / r, which also falls as r grows.

Field lines point from higher potential to lower potential, so moving along a field line means moving to lower potential.

12.3 How can you identify that which plate of a capacitor is positively charged?

The positively charged plate is the one at the higher potential.

Electric field lines start on the positive plate and end on the negative plate.

A small positive test charge placed between the plates is pushed away from the positive plate and pulled towards the negative plate.

In a circuit, it is the plate joined to the positive terminal of the battery.

12.4 Describe the force or forces on a positive point charge when placed between parallel plates (a) with similar and equal charges (b) with opposite and equal charges.

(a) Similar and equal charges: the field of one plate cancels the field of the other between the plates.

The net field between the plates is zero, so no force acts on the charge there.

(b) Opposite and equal charges: the fields of the two plates add between the plates.

The result is a uniform field pointing from the positive plate to the negative plate.

The charge feels a force F = qE towards the negative plate.

12.5 Electric lines of force never cross. Why?

An electric line of force shows the direction of the field at every point on it.

The field at any one point has a single, unique direction.

If two lines crossed, the field at the crossing point would have two directions at once.

That is impossible, so lines of force never cross.

12.6 If a point charge q of mass m is released in a non-uniform electric field with field lines pointing in the same direction, will it make a rectilinear motion?

Yes, its motion is rectilinear, that is, along a straight line.

The force on the charge is always along the field line, F = qE, and the lines all point the same way.

So the charge keeps moving along that straight direction.

Because the field is non-uniform, the size of the force, and hence the acceleration, changes along the path, but the direction does not.

12.7 Is E necessarily zero inside a charged rubber balloon if balloon is spherical? Assume that charge is distributed uniformly over the surface.

Yes, the field inside is zero everywhere.

Charge on a conductor, such as the charged rubber surface, stays on the outer surface.

By symmetry, and by Gauss's law for a spherical shell, the fields of all the surface charges cancel at any inside point.

So E = 0 inside the uniformly charged spherical balloon.

12.8 Is it true that Gauss's law states that the total number of lines of force crossing any closed surface in the outward direction is proportional to the net positive charge enclosed within surface?

Yes, this statement is true.

Gauss's law says the total electric flux through a closed surface equals the enclosed

charge divided by epsilon-zero.

The number of lines of force crossing the surface is a picture of that flux.

So the outward lines are proportional to the net positive charge enclosed.

12.9 Do electrons tend to go to region of high potential or of low potential?

Electrons tend to go to the region of high potential.

The field points from high potential to low potential.

The force on a negative charge is opposite to the field, F = -eE.

So an electron is pushed towards higher potential, against the field direction.

Exercise - PROBLEMS 12.1 to 12.13 (book pp. 28-30, verbatim, with the book's printed answers)

Problem 12.1 - Compare magnitudes of electrical and gravitational forces exerted on an object (mass = 10.0 g, charge = 20.0 microC) by an identical object placed 10.0 cm from the first. (G = 6.67 x 10^-11 N m^2 kg^-2) (Book answer: Fe/Fg = 5.4 x 10^14)

Convert to SI units: m = 10.0 g = 0.010 kg, q = 20.0 microC = 20.0 x 10^-6 C, r = 10.0 cm = 0.10 m.

Electrical force: Fe = k q squared / r squared.

Fe = (9.0 x 10^9) x (20.0 x 10^-6) squared / (0.10) squared.

(20.0 x 10^-6) squared = 400 x 10^-12 = 4.0 x 10^-10.

Fe = (9.0 x 10^9) x (4.0 x 10^-10) / 0.010 = 3.6 / 0.010 = 360 N.

Gravitational force: Fg = G m squared / r squared.

Fg = (6.67 x 10^-11) x (0.010) squared / (0.10) squared.

Fg = (6.67 x 10^-11) x (1.0 x 10^-4) / 0.010 = 6.67 x 10^-13 N.

Ratio: Fe / Fg = 360 / (6.67 x 10^-13) = 5.4 x 10^14.

Answer: Fe / Fg = 5.4 x 10^14, as printed in the book.

Problem 12.2 - Calculate vectorially the net electrostatic force on q as shown in the figure. Charges: q = 4.0 microC at the top, q1 = 1.0 microC and q2 = -1.0 microC at the base corners; each slant side 1.0 m, base halves 0.60 m, height 0.80 m. (Book answer: F = 0.058 i-hat N - see note)

Step 1: Force on q due to q1. Both are positive, so it is repulsion, away from q1, along the left slant.

Magnitude: F1 = k q q1 / r squared = (9.0 x 10^9)(4.0 x 10^-6)(1.0 x 10^-6) / (1.0) squared.

F1 = 0.036 N.

Step 2: Resolve F1. From q1 to q the displacement is (0.60 m across, 0.80 m up) over 1.0 m, so the unit vector is (0.60, 0.80).

F1 vector = 0.036 x (0.60 i-hat + 0.80 j-hat) = 0.0216 i-hat + 0.0288 j-hat N.

Step 3: Force on q due to q2. q2 is negative, so it is attraction, towards q2, along the right slant.

Magnitude: F2 = k q |q2| / r squared = 0.036 N, the same as F1.

Step 4: Resolve F2. From q to q2 the displacement is (0.60 m across, 0.80 m down) over 1.0 m, so the unit vector is (0.60, -0.80).

F2 vector = 0.036 x (0.60 i-hat - 0.80 j-hat) = 0.0216 i-hat - 0.0288 j-hat N.

Step 5: Add the vectors. The j-hat parts cancel: 0.0288 - 0.0288 = 0.

The i-hat parts add: 0.0216 + 0.0216 = 0.0432.

Component-wise result: F = 0.043 i-hat N.

NOTE ON THE PRINTED ANSWER: the book prints F = 0.058 i-hat N, which is 2 x 0.036 x 0.80 = 0.0576 N, that is, it uses 0.80 where the figure's x-component ratio is 0.60 / 1.0 = 0.60. The step-by-step addition from the figure's own numbers gives 0.043 i-hat N; the book's printed 0.058 i-hat N is recorded here as printed, not silently substituted.

Problem 12.3 - A point charge q = -8.0 x 10^-8 C is placed at the origin. Calculate electric field at a point 2.0 m from the origin on the z-axis. [Book answer: (-1.8 x 10^2 k-hat) N C^-1]

Field of a point charge: E = k q / r squared, in the direction away from a positive charge and towards a negative charge.

Magnitude: |E| = (9.0 x 10^9)(8.0 x 10^-8) / (2.0) squared.

|E| = 720 / 4.0 = 180 = 1.8 x 10^2 N C^-1.

The charge is negative, so at a point on the +z-axis the field points back towards the origin, that is, in the -k-hat direction.

Answer: E = (-1.8 x 10^2 k-hat) N C^-1, as printed in the book.

Problem 12.4 - Determine the electric field at the position r = (4 i-hat + 3 j-hat) m caused by a point charge q = 5.0 x 10^-6 C placed at origin. [Book answer: (1440 i-hat + 1080 j-hat) N C^-1]

Distance from origin: r = sqrt(4 squared + 3 squared) = sqrt(25) = 5.0 m.

Unit vector along r: r-hat = (4 i-hat + 3 j-hat) / 5 = 0.80 i-hat + 0.60 j-hat.

Field magnitude: E = k q / r squared = (9.0 x 10^9)(5.0 x 10^-6) / (5.0) squared.

E = 45000 / 25 = 1800 N C^-1.

Vector: E = 1800 x (0.80 i-hat + 0.60 j-hat).

E = 1440 i-hat + 1080 j-hat N C^-1.

Answer: (1440 i-hat + 1080 j-hat) N C^-1, as printed in the book.

Problem 12.5 - Two point charges, q1 = -1.0 x 10^-6 C and q2 = +4.0 x 10^-6 C, are separated by a distance of 3.0 m. Find and justify the zero-field location. (Book answer: 3.0 m)

The zero-field point cannot lie between the charges, because there the two fields point the same way and add.

It cannot lie beyond the larger charge q2, because q2 always dominates there.

It must lie beyond the smaller charge q1, on the side away from q2.

Let its distance from q1 be x. Its distance from q2 is then x + 3.0 m.

Set the magnitudes equal: k |q1| / x squared = k |q2| / (x + 3.0) squared.

Cancel k and take square roots: (x + 3.0) / x = sqrt(|q2| / |q1|) = sqrt(4.0) = 2.

So x + 3.0 = 2x, giving x = 3.0 m.

Answer: 3.0 m from q1, on the side away from q2, as printed in the book.

Problem 12.6 - Find the electric field strength required to hold suspended a particle of mass 1.0 x 10^-6 kg and charge 1.0 microC between two plates 10.0 cm apart. (Book answer: 9.8 V m^-1)

For suspension, the upward electric force must balance the weight: qE = mg.

So E = mg / q.

E = (1.0 x 10^-6 kg)(9.8 m s^-2) / (1.0 x 10^-6 C).

E = 9.8 x 10^-6 / 1.0 x 10^-6 = 9.8 V m^-1.

Answer: 9.8 V m^-1, as printed in the book. (The 10.0 cm plate separation is not needed for E itself; it would give the plate voltage V = E d = 0.98 V.)

Problem 12.7 - A particle having a charge of 20 electrons on it falls through a potential difference of 100 volts. Calculate the energy acquired by it in electron volts (eV). [Book answer: (2.0 x 10^3 eV)]

Charge on the particle: q = 20 e.

Energy gained falling through potential V: Energy = qV.

Energy = 20 e x 100 V = 2000 eV.

Answer: 2.0 x 10^3 eV, as printed in the book.

Problem 12.8 - Millikan's experiment: oil droplets between two flat horizontal plates, 5.00 mm apart. Plate voltage adjusted to exactly 780 V so the droplet is held stationary. Voltage switched off; the droplet falls 1.50 mm in 11.2 s. Density of oil 900 kg m^-3, viscosity of air 1.80 x 10^-5 N s m^-2. Calculate (a) the mass, (b) the charge on the droplet (g = 9.8 m s^-2). [Book answers: (a) 5.14 x 10^-15 kg, (b) 3.20 x 10^-19 C]

Step 1: Terminal velocity of fall: v = distance / time = 1.50 x 10^-3 m / 11.2 s = 1.34 x 10^-4 m s^-1.

Step 2: At terminal velocity the Stokes drag balances the weight: 6 pi eta r v = mg, with m = (4/3) pi r cubed x density.

Step 3: Solve for the radius: r squared = 9 eta v / (2 x density x g).

r squared = 9 x (1.80 x 10^-5) x (1.34 x 10^-4) / (2 x 900 x 9.8).

r squared = 2.17 x 10^-8 / 17640 = 1.23 x 10^-12, so r = 1.11 x 10^-6 m.

Step 4: Mass: m = (4/3) pi r cubed x density = (4/3) pi (1.11 x 10^-6) cubed x 900.

m = 5.14 x 10^-15 kg. Answer (a) matches the book.

Step 5: When held stationary, electric force balances weight: qE = mg, with E = V / d.

E = 780 V / 5.00 x 10^-3 m = 1.56 x 10^5 V m^-1.

Step 6: q = mg / E = (5.14 x 10^-15)(9.8) / (1.56 x 10^5).

q = 5.04 x 10^-14 / 1.56 x 10^5 = 3.20 x 10^-19 C. Answer (b) matches the book.

The charge is twice the elementary charge 1.60 x 10^-19 C, as expected for a quantized charge.

Problem 12.9 - A proton placed in a uniform electric field of 5000 N C^-1 directed to right is allowed to go a distance of 10.0 cm from A to B. Calculate (a) potential difference between the two points (b) work done by the field (c) change in P.E. of proton (d) change in K.E. of the proton (e) its velocity (mass of proton 1.67 x 10^-27 kg). (Book answers: -500 V, 500 eV, -500 eV, 500 eV, 3.097 x 10^5 m s^-1)

Step 1: (a) Moving along the field, potential falls: V(B) - V(A) = -E d.

V(B) - V(A) = -(5000)(0.10) = -500 V.

Step 2: (b) Work done by the field: W = q E d = (1.60 x 10^-19 C)(5000 N C^-1)(0.10 m).

W = 8.0 x 10^-17 J. In electron volts: 8.0 x 10^-17 / 1.60 x 10^-19 = 500 eV.

Step 3: (c) Change in potential energy is the negative of the work done by the field.

Change in P.E. = -500 eV.

Step 4: (d) By energy conservation, the lost potential energy becomes kinetic energy.

Change in K.E. = +500 eV.

Step 5: (e) (1/2) m v squared = 8.0 x 10^-17 J.

v squared = 2 x (8.0 x 10^-17) / (1.67 x 10^-27) = 9.58 x 10^10.

v = 3.097 x 10^5 m s^-1.

Answers: -500 V, 500 eV, -500 eV, 500 eV, 3.097 x 10^5 m s^-1, as printed in the book.

Problem 12.10 - Using zero reference point at infinity, determine the amount by which a point charge of 4.0 x 10^-8 C alters the electric potential at a point 1.2 m

away, when (a) charge is positive (b) charge is negative. (Book answers: +3.0 x 10^2 V, -3.0 x 10^2 V)

Potential of a point charge with zero at infinity: V = k q / r.

(a) Positive charge: V = (9.0 x 10^9)(4.0 x 10^-8) / 1.2.

V = 360 / 1.2 = 300 = +3.0 x 10^2 V.

(b) Negative charge: q = -4.0 x 10^-8 C, so V = -300 V.

Answers: +3.0 x 10^2 V and -3.0 x 10^2 V, as printed in the book.

Problem 12.11 - In Bohr's atomic model of hydrogen atom, the electron is in an orbit around the nuclear proton at a distance of 5.29 x 10^-11 m with a speed of 2.18 x 10^6 m s^-1 (e = 1.60 x 10^-19 C, mass of electron = 9.10 x 10^-31 kg). Find (a) the electric potential that a proton exerts at this distance (b) total energy of the atom in eV (c) the ionization energy for the atom in eV. (Book answers: +27.20 V, -13.6 eV, +13.60 eV)

Step 1: (a) Potential due to the proton: V = k e / r.

V = (9.0 x 10^9)(1.60 x 10^-19) / (5.29 x 10^-11).

V = 1.44 x 10^-9 / 5.29 x 10^-11 = 27.20 V.

Step 2: (b) Kinetic energy of the electron: K.E. = (1/2) m v squared.

K.E. = 0.5 x (9.10 x 10^-31) x (2.18 x 10^6) squared.

K.E. = 0.5 x (9.10 x 10^-31) x (4.75 x 10^12) = 2.16 x 10^-18 J.

In eV: 2.16 x 10^-18 / 1.60 x 10^-19 = 13.5 eV.

Step 3: Potential energy: P.E. = -eV = -(1.60 x 10^-19 C)(27.20 V) = -27.2 eV.

Step 4: Total energy = K.E. + P.E. = 13.5 eV - 27.2 eV = -13.6 eV (using the book's rounded values, -13.6 eV).

Step 5: (c) Ionization energy is the energy needed to free the electron, the negative of the total energy.

Ionization energy = +13.60 eV.

Answers: +27.20 V, -13.6 eV, +13.60 eV, as printed in the book.

Problem 12.12 - The electronic flash attachment for a camera contains a capacitor for storing the energy used to produce the flash. In one such unit, the potential difference between the plates of a 750 microF capacitor is 330 V. Determine the energy that is used to produce the flash. (Book answer: 40.8 J)

Energy stored in a capacitor: U = (1/2) C V squared.

C = 750 microF = 750 x 10^-6 F, V = 330 V.

U = 0.5 x (750 x 10^-6) x (330) squared.

(330) squared = 108900.

U = 0.5 x (750 x 10^-6) x 108900 = 0.5 x 81.675 = 40.8 J.

Answer: 40.8 J, as printed in the book.

Problem 12.13 - A capacitor has a capacitance of 2.5 x 10^-8 F. In the charging process, electrons are removed from one plate and placed on the other one. When the potential difference between the plates is 450 V, how many electrons have been transferred? (e = 1.60 x 10^-19 C) (Book answer: 7.0 x 10^13 electrons)

Charge stored: Q = C V.

Q = (2.5 x 10^-8 F)(450 V) = 1.125 x 10^-5 C.

Number of electrons: n = Q / e.

n = (1.125 x 10^-5) / (1.60 x 10^-19) = 7.03 x 10^13.

Answer: 7.0 x 10^13 electrons, as printed in the book.

Chapter 13: Current Electricity

Exercise - QUESTIONS 13.1 to 13.9 (book pp. 53-54, verbatim)

13.1 A potential difference is applied across the ends of a copper wire. What is the effect on the drift velocity of free electrons by (i) increasing the potential difference (ii) decreasing the length and the temperature of the wire

The drift velocity is v = I / (n A e), and it is proportional to the electric field in the

wire, v is proportional to E = V / L.

(i) Increasing the potential difference: the field E = V / L increases.

The electrons are driven harder, so the drift velocity increases.

(ii) Decreasing the length: at the same potential difference, E = V / L increases because L is smaller.

So the drift velocity increases.

Decreasing the temperature: the metal ions vibrate less, so the electrons collide less often and move more freely.

The resistance falls and the drift velocity increases.

So in both cases the drift velocity increases.

13.2 Do bends in a wire affect its electrical resistance? Explain.

No, bends in a wire do not appreciably affect its resistance.

The resistance of a wire is R = rho L / A, where L is the length measured along the wire, A is its cross-sectional area, and rho is the resistivity of the material.

Bending the wire changes neither its length along the wire, nor its cross-sectional area, nor its material.

So the resistance stays the same; only the shape of the path changes.

13.3 What are the resistances of the resistors given in the figures A and B? What is the tolerance of each? Explain what is meant by the tolerance? (Fig. Q. 13.3: A: Brown, Green, Red, Gold; B: Yellow, White, Orange, Silver)

Colour code: Black 0, Brown 1, Red 2, Orange 3, Yellow 4, Green 5, Blue 6, Violet 7, Grey 8, White 9. The first two bands are digits, the third is the multiplier, the fourth is the tolerance.

Resistor A: Brown (1), Green (5), Red (multiplier 10 squared), Gold (tolerance).

Resistance = 15 x 10 squared = 1500 ohm = 1.5 kilo-ohm, tolerance 5 percent (Gold).

Resistor B: Yellow (4), White (9), Orange (multiplier 10 cubed), Silver (tolerance).

Resistance = 49 x 10 cubed = 49000 ohm = 49 kilo-ohm, tolerance 10 percent (Silver).

Tolerance means the permitted variation of the actual resistance from the marked value.

For example, a 1500 ohm resistor with 5 percent tolerance may actually be anywhere from 1425 ohm to 1575 ohm.

13.4 Why does the resistance of a conductor rise with temperature?

In a metal, current is carried by free electrons moving through a lattice of vibrating ions.

When the temperature rises, the ions vibrate with larger amplitude.

The free electrons then collide more often with the ions and lose more of their drift energy.

More collisions mean more opposition to the flow of charge.

So the resistivity, and hence the resistance, of the conductor rises with temperature.

13.5 What are the difficulties in testing whether the filament of a lighted bulb obeys Ohm's law?

Ohm's law, V = IR with R constant, holds only if the physical state, especially the temperature, stays constant.

A lighted bulb filament does not stay at one temperature: the current itself heats it, and its temperature rises far above room temperature.

As it heats, its resistance rises greatly, so the ratio V / I is not constant.

The filament is therefore non-ohmic in use, and a test at one steady temperature is hard to arrange.

Measuring the cold resistance and the hot resistance separately, and keeping the filament at a fixed temperature while varying V, are the practical difficulties.

13.6 Is the filament resistance lower or higher in a 500 W, 220 V light bulb than in a

100 W, 220 V bulb?

For a bulb, power P = V squared / R, so R = V squared / P.

For the 500 W bulb: R = (220) squared / 500 = 48400 / 500 = 96.8 ohm.

For the 100 W bulb: R = (220) squared / 100 = 48400 / 100 = 484 ohm.

The 500 W bulb draws more current at the same voltage, so it must have the smaller resistance.

Answer: the filament resistance is LOWER in the 500 W bulb (about one-fifth of the 100 W bulb's).

13.7 Describe a circuit which will give a continuously varying potential.

A potential divider (rheostat) circuit gives a continuously varying potential.

Connect a rheostat, a long uniform resistance wire with a sliding contact, across the terminals of a battery.

Take the output between one fixed end of the rheostat and the sliding contact.

As the contact slides along the wire, the length of wire in the output part changes smoothly.

The output potential therefore varies continuously from zero up to the full battery voltage.

13.8 Explain why the terminal potential difference of a battery decreases when the current drawn from it is increased?

The terminal potential difference is V = E - I r, where E is the emf and r is the internal resistance of the battery.

When the battery supplies current, part of its emf is used up in driving the current through its own internal resistance.

This lost voltage is I r, and it grows as the current I grows.

So the larger the current drawn, the larger the internal drop, and the smaller the terminal potential difference V.

13.9 What is Wheatstone bridge? How can it be used to determine an unknown

resistance?

A Wheatstone bridge is a network of four resistors arranged in a diamond (two parallel branches, each of two resistors in series), with a battery across one diagonal and a galvanometer across the other diagonal.

The bridge is balanced when the galvanometer shows no deflection, meaning no current flows through it.

At balance, the ratio of the resistances in one branch equals the ratio in the other branch, P / Q = R / S.

If three resistances are known and the fourth is unknown, the unknown is found from this balance condition.

For example, if the unknown is S, then S = R x (Q / P), using the three known resistors.

Exercise - PROBLEMS 13.1 to 13.8 (book pp. 54-55, verbatim, with the book's printed answers)

Problem 13.1 - How many electrons pass through an electric bulb in one minute if the 300 mA current is passing through it? (Book answer: 1.12 x 10^20)

Current: I = 300 mA = 0.30 A. Time: t = 1 minute = 60 s.

Charge passed: Q = I t.

Q = 0.30 x 60 = 18 C.

Number of electrons: n = Q / e, with e = 1.60 x 10^-19 C.

n = 18 / (1.60 x 10^-19) = 1.125 x 10^20.

Answer: 1.12 x 10^20 electrons, as printed in the book.

Problem 13.2 - A charge of 90 C passes through a wire in 1 hour and 15 minutes. What is the current in the wire? (Book answer: 20 mA)

Charge: Q = 90 C.

Time: t = 1 hour 15 minutes = 75 minutes = 75 x 60 = 4500 s.

Current: I = Q / t.

I = 90 / 4500 = 0.020 A.

Answer: 20 mA, as printed in the book.

Problem 13.3 - Find the equivalent resistance of the circuit (Fig. P. 13.3), total current drawn from the source and the current through each resistor. (Fig: R1 = 6 ohm and R2 = 6 ohm in parallel, in series with R3 = 3 ohm; source 6 V.) (Book answer: 6.0 ohm, 1.0 A, 0.5 A, 0.5 A, 1.0 A)

Step 1: Parallel part. R1 and R2 are in parallel: 1 / Rp = 1 / 6 + 1 / 6 = 2 / 6.

So Rp = 3.0 ohm.

Step 2: Equivalent resistance. Rp is in series with R3: Req = 3.0 + 3.0 = 6.0 ohm.

Step 3: Total current from the source: I = V / Req = 6 / 6.0 = 1.0 A.

Step 4: The total current divides equally between the two equal parallel resistors.

Current through R1 = 0.5 A, current through R2 = 0.5 A.

Step 5: The full total current flows through R3, which is in series with the combination.

Current through R3 = 1.0 A.

Answers: 6.0 ohm, 1.0 A, 0.5 A, 0.5 A, 1.0 A, as printed in the book.

Problem 13.4 - A rectangular bar of iron is 2.0 cm by 2.0 cm in cross section and 40 cm long. Calculate its resistance if the resistivity of iron is 11 x 10^-8 ohm m. (Book answer: 1.1 x 10^-4 ohm)

Cross-sectional area: A = 2.0 cm x 2.0 cm = 4.0 cm squared = 4.0 x 10^-4 m squared.

Length: L = 40 cm = 0.40 m. Resistivity: rho = 11 x 10^-8 ohm m.

Resistance: R = rho L / A.

R = (11 x 10^-8) x 0.40 / (4.0 x 10^-4).

R = (4.4 x 10^-8) / (4.0 x 10^-4) = 1.1 x 10^-4 ohm.

Answer: 1.1 x 10^-4 ohm, as printed in the book.

Problem 13.5 - The resistance of an iron wire at 0 degrees C is 1 x 10^4 ohm. What is the resistance at 500 degrees C if the temperature coefficient of resistance of iron

is 5.2 x 10^-3 K^-1? (Book answer: 3.6 x 10^4 ohm)

Resistance at temperature t: Rt = R0 (1 + alpha t).

Here R0 = 1 x 10^4 ohm, alpha = 5.2 x 10^-3 K^-1, t = 500 degrees C.

alpha t = (5.2 x 10^-3) x 500 = 2.6.

Rt = (1 x 10^4) x (1 + 2.6) = (1 x 10^4) x 3.6.

Answer: 3.6 x 10^4 ohm, as printed in the book.

Problem 13.6 - Calculate terminal potential difference of each of cells in circuit of Fig. P. 13.6. (Fig: 24 V cell with internal resistance 0.10 ohm, 6.0 V cell with internal resistance 0.90 ohm, in a loop with an 8.0 ohm resistor.) (Book answers: 23.8 V, 7.8 V)

Step 1: The two cells oppose each other in the loop, so the net emf is 24 - 6.0 = 18 V.

Step 2: Total resistance in the loop: 0.10 + 8.0 + 0.90 = 9.0 ohm.

Step 3: Loop current: I = 18 / 9.0 = 2.0 A.

Step 4: The 24 V cell is discharging, so its terminal potential difference is V = E - I r.

V = 24 - (2.0)(0.10) = 24 - 0.20 = 23.8 V.

Step 5: The 6.0 V cell is being charged by the larger cell, so its terminal potential

difference is V = E + I r.

V = 6.0 + (2.0)(0.90) = 6.0 + 1.8 = 7.8 V.

Answers: 23.8 V and 7.8 V, as printed in the book.

Problem 13.7 - Find the current which flows in all the resistances of the circuit of Fig. P. 13.7. (Fig: 9.0 V cell in the left branch, 18 ohm resistor in the middle branch, 6.0 V cell in series with a 12 ohm resistor in the right branch.) (Book answers: 1.25 A, 0.5 A)

Step 1: The left branch is a 9.0 V cell alone, so the potential difference across the middle branch is 9.0 V.

Step 2: Current through the 18 ohm middle resistor: I = V / R = 9.0 / 18 = 0.5 A.

Step 3: In the right loop the two cells drive current the same way round, so the net emf is 9.0 + 6.0 = 15 V.

Step 4: Current through the 12 ohm resistor: I = 15 / 12 = 1.25 A.

Answers: 1.25 A through the 12 ohm resistor and 0.5 A through the 18 ohm resistor, as printed in the book.

Problem 13.8 - Find the current and power dissipated in each resistance of the circuit, shown in Fig. P. 13.8. (Fig: left branch 6.0 V cell; top-left 1.0 ohm, middle 2.0 ohm, bottom-left 1.0 ohm; top-right 1.0 ohm, right 2.0 ohm, bottom-right 1.0 ohm; right branch 10 V cell.) (Book answers: 0.8 A, 1.4 A, 2.2 A, 0.64 W, 1.96 W, 3.92 W,

9.68 W)

Step 1: Let I1 be the current in the left loop and I2 the current in the right loop. At the middle junction they add, so the middle 2.0 ohm resistor carries I1 + I2.

Step 2: Left loop (6.0 V cell, two 1.0 ohm resistors, middle 2.0 ohm resistor): 6.0 = I1 (1.0 + 1.0) + 2.0 (I1 + I2).

So 6.0 = 4 I1 + 2 I2. (Equation 1)

Step 3: Right loop (10 V cell, two 1.0 ohm resistors, one 2.0 ohm resistor, middle 2.0 ohm resistor): 10 = I2 (1.0 + 2.0 + 1.0) + 2.0 (I1 + I2).

So 10 = 2 I1 + 6 I2. (Equation 2)

Step 4: Solve. Multiply Equation 2 by 2: 20 = 4 I1 + 12 I2. Subtract Equation 1: 14 = 10 I2, so I2 = 1.4 A.

Step 5: From Equation 1: 4 I1 = 6.0 - 2 (1.4) = 6.0 - 2.8 = 3.2, so I1 = 0.8 A.

Step 6: Middle current: I1 + I2 = 0.8 + 1.4 = 2.2 A.

Step 7: Powers, P = I squared R. Left 1.0 ohm resistors: (0.8) squared x 1.0 = 0.64 W each.

Right 1.0 ohm resistors: (1.4) squared x 1.0 = 1.96 W each.

Right 2.0 ohm resistor: (1.4) squared x 2.0 = 3.92 W.

Middle 2.0 ohm resistor: (2.2) squared x 2.0 = 9.68 W.

Answers: 0.8 A, 1.4 A, 2.2 A, 0.64 W, 1.96 W, 3.92 W, 9.68 W, as printed in the book.

Chapter 14: Electromagnetism

Exercise - QUESTIONS 14.1 to 14.15 (book p. 79, verbatim)

14.1 A plane conducting loop is located in a uniform magnetic field that is directed along the x-axis. For what orientation of the loop is the flux a maximum? For what orientation is the flux a minimum?

Magnetic flux through the loop is phi = B A cos theta, where theta is the angle between the field and the normal to the plane of the loop.

The flux is a maximum when cos theta = 1, that is, theta = 0.

This happens when the normal to the loop is along the field, so the plane of the loop is perpendicular to the x-axis.

The flux is a minimum, zero, when cos theta = 0, that is, theta = 90 degrees.

This happens when the normal to the loop is perpendicular to the field, so the plane of the loop is parallel to the x-axis.

14.2 A current in a conductor produces a magnetic field, which can be calculated using Ampere's law. Since current is defined as the rate of flow of charge, what can you conclude about the magnetic field due to stationary charges? What about moving charges?

Ampere's law links the magnetic field to the current enclosed, and current is moving charge.

Stationary charges have no current, so they produce no magnetic field at all.

Stationary charges produce only an electric field.

Moving charges, that is, a current, do produce a magnetic field around them.

So magnetism is produced by moving charges, not by charges at rest.

14.3 Describe the change in the magnetic field inside a solenoid carrying a steady current I, if (a) the length of the solenoid is doubled but the number of turns remains the same and (b) the number of turns is doubled, but the length remains the same.

The field inside a solenoid is B = mu-zero n I, where n = N / L is the number of turns per unit length.

(a) Length doubled, turns unchanged: n becomes N / (2L), half its old value.

So the magnetic field is halved.

(b) Turns doubled, length unchanged: n becomes 2N / L, twice its old value.

So the magnetic field is doubled.

14.4 At a given instant, a proton moves in the positive x direction in a region where there is magnetic field in the negative z direction. What is the direction of the magnetic force? Will the proton continue to move in the positive x direction? Explain.

The magnetic force is F = q v cross B. For a proton q is positive.

Velocity is along +x and the field is along -z, so v cross B is along +y, since x-hat cross (-z-hat) = +y-hat.

So the force on the proton is in the positive y direction.

No, the proton will not continue in the positive x direction.

The force is always perpendicular to its velocity, so it deflects the proton sideways.

The proton moves in a circular path in the x-y plane, its speed staying constant while its direction keeps changing.

14.5 Two charged particles are projected into a region where there is a magnetic

field perpendicular to their velocities. If the charges are deflected in opposite directions, what can you say about them?

The magnetic force is F = q v cross B, so its direction depends on the sign of the charge.

Two particles entering with the same velocity direction in the same field feel forces in opposite directions only if their charges have opposite signs.

So one particle is positively charged and the other is negatively charged.

The opposite deflection is the signature of opposite charges.

14.6 Suppose that a charge q is moving in a uniform magnetic field with a velocity v. Why is there no work done by the magnetic force that acts on the charge q?

The magnetic force F = q v cross B is always perpendicular to the velocity v.

The displacement of the charge in any short time is along its velocity.

Work is F dot displacement, and the dot product of perpendicular vectors is zero.

So the magnetic force does no work on the charge.

It changes the direction of motion, but never the speed or the kinetic energy.

14.7 If a charge particle moves in a straight line through some region of space, can you say that the magnetic field in the region is zero?

No, a straight-line path does not prove the field is zero.

If the particle moves parallel to the magnetic field, the angle between v and B is zero, so the magnetic force q v B sin theta is zero.

The particle then goes straight even though a field is present.

Also, an uncharged particle would go straight through any field.

So a straight path alone cannot prove that B is zero.

14.8 Why does the picture on a TV screen become distorted when a magnet is brought near the screen?

A TV picture is made by a beam of electrons striking the screen, and the beam is steered by magnetic fields inside the set.

A magnet brought near the screen adds its own magnetic field in the region of the beam.

This extra field deflects the electrons away from their intended paths.

Electrons then land at the wrong places on the screen, so the picture and colours become distorted.

14.9 Is it possible to orient a current loop in a uniform magnetic field such that the loop will not tend to rotate? Explain.

Yes, it is possible. The torque on a current loop is tau = N I A B sin theta, where theta is the angle between the normal to the loop and the field.

The torque is zero when sin theta = 0, that is, when the normal to the loop is along the field.

This means the plane of the loop is perpendicular to the magnetic field.

In this orientation the loop is in stable equilibrium and has no tendency to rotate.

14.10 How can a current loop be used to determine the presence of a magnetic field in a given region of space?

Place a small current-carrying loop, suspended so it can turn freely, in the region.

If a magnetic field is present, the loop experiences a torque and turns until its plane is perpendicular to the field.

If the loop shows no tendency to turn in any orientation, there is no magnetic field in the region.

The direction in which the loop settles also shows the direction of the field.

So the loop acts as a detector of both the presence and the direction of the field.

14.11 How can you use a magnetic field to separate isotopes of chemical element?

Isotopes of an element have the same charge but different masses.

Give the ions of the element the same kinetic energy, for example by accelerating them through the same potential difference.

Then send them into a uniform magnetic field perpendicular to their motion.

Each ion follows a circular path of radius r = m v / (q B), which depends on its mass.

Heavier isotopes follow larger circles and lighter ones smaller circles, so the isotopes land at different places and are separated.

14.12 What should be the orientation of a current carrying coil in a magnetic field so that torque acting upon the coil is (a) maximum (b) minimum?

The torque on a coil is tau = N I A B sin theta, where theta is the angle between the normal to the coil and the field.

(a) Maximum: sin theta = 1, theta = 90 degrees, so the normal to the coil is perpendicular to the field.

This means the plane of the coil is parallel to the magnetic field.

(b) Minimum: sin theta = 0, theta = 0, so the normal to the coil is along the field.

This means the plane of the coil is perpendicular to the magnetic field, and the torque is zero.

14.13 A loop of wire is suspended between the poles of a magnet with its plane parallel to the pole faces. What happens if a direct current is put through the coil? What happens if an alternating current is used instead?

With the plane parallel to the pole faces, the plane is parallel to the field, so the torque on the loop is at its maximum.

A direct current gives a steady torque in one direction, so the loop rotates until its plane is perpendicular to the field, where the torque becomes zero, and it settles there.

An alternating current reverses direction every half cycle, so the torque also reverses every half cycle.

The loop therefore cannot rotate steadily; instead it vibrates or oscillates about its mean position.

14.14 Why the resistance of an ammeter should be very low?

An ammeter is connected in series with the circuit whose current is to be measured.

If its resistance were large, it would reduce the current it is trying to measure and change the circuit.

A very low resistance ammeter takes almost no voltage and hardly changes the current.

So the current it reads is very close to the true current in the circuit.

14.15 Why the voltmeter should have a very high resistance?

A voltmeter is connected in parallel across the component whose potential difference is to be measured.

If its resistance were low, it would draw a large current and change the potential difference it is trying to measure.

A very high resistance voltmeter draws almost no current from the circuit.

So it measures the potential difference without disturbing the circuit.

Exercise - PROBLEMS 14.1 to 14.10 (book p. 80, verbatim, with the book's printed answers)

Problem 14.1 - Find the value of the magnetic field that will cause a maximum force of 7.0 x 10^-3 N on a 20.0 cm straight wire carrying a current of 10.0 A. (Book answer: 3.5 x 10^-3 T)

Maximum force on a current-carrying wire: F = I L B (when the wire is perpendicular to the field).

So B = F / (I L).

L = 20.0 cm = 0.20 m, I = 10.0 A, F = 7.0 x 10^-3 N.

B = (7.0 x 10^-3) / (10.0 x 0.20) = (7.0 x 10^-3) / 2.0.

Answer: 3.5 x 10^-3 T, as printed in the book.

Problem 14.2 - How fast must a proton move in a magnetic field of 2.50 x 10^-3 T such that the magnetic force is equal to its weight? (Book answer: 4.09 x 10^-5 m s^-1)

Magnetic force on the proton: F = q v B (motion perpendicular to the field).

Weight of the proton: W = m g.

Set them equal: q v B = m g, so v = m g / (q B).

Mass of proton m = 1.67 x 10^-27 kg, q = 1.60 x 10^-19 C, g = 9.8 m s^-2, B = 2.50 x 10^-3 T.

v = (1.67 x 10^-27)(9.8) / ((1.60 x 10^-19)(2.50 x 10^-3)).

v = (1.64 x 10^-26) / (4.0 x 10^-22) = 4.09 x 10^-5 m s^-1.

Answer: 4.09 x 10^-5 m s^-1, as printed in the book.

Problem 14.3 - A velocity selector has a magnetic field of 0.30 T. If a perpendicular electric field of 10,000 V m^-1 is applied, what will be the speed of the particle that will pass through the selector? (Book answer: 3.3 x 10^4 m s^-1)

In a velocity selector the electric and magnetic forces balance: q E = q v B.

So v = E / B.

v = 10000 / 0.30 = 33333 m s^-1.

Answer: 3.3 x 10^4 m s^-1, as printed in the book.

Problem 14.4 - A coil of 0.1 m x 0.1 m and of 200 turns carrying a current of 1.0 mA is placed in a uniform magnetic field of 0.1 T. Calculate the maximum torque that acts on the coil. (Book answer: 2.0 x 10^-4 N m)

Area of the coil: A = 0.1 x 0.1 = 0.01 m squared.

Maximum torque: tau = N I A B (when sin theta = 1).

tau = 200 x (1.0 x 10^-3) x 0.01 x 0.1.

tau = 200 x 1.0 x 10^-6 = 2.0 x 10^-4 N m.

Answer: 2.0 x 10^-4 N m, as printed in the book.

Problem 14.5 - A power line 10.0 m high carries a current 200 A. Find the magnetic field of the wire at the ground. (Book answer: 4.0 x 10^-6 T)

Field of a long straight wire: B = (mu-zero I) / (2 pi r).

Here I = 200 A and r = 10.0 m (the height of the line above the ground).

B = (4 pi x 10^-7 x 200) / (2 pi x 10.0).

B = (2 x 10^-7 x 200) / 10.0 = 4.0 x 10^-6 T.

Answer: 4.0 x 10^-6 T, as printed in the book.

Problem 14.6 - You are asked to design a solenoid that will give a magnetic field of 0.10 T, yet the current must not exceed 10.0 A. Find the number of turns per unit length that the solenoid should have. (Book answer: 7.96 x 10^3)

Field of a solenoid: B = mu-zero n I, where n is turns per unit length.

So n = B / (mu-zero I).

n = 0.10 / ((4 pi x 10^-7) x 10.0).

n = 0.10 / (1.256 x 10^-5) = 7960.

Answer: 7.96 x 10^3 turns per metre, as printed in the book.

Problem 14.7 - What current should pass through a solenoid that is 0.5 m long with 10,000 turns of copper wire so that it will have a magnetic field of 0.4 T? (Book answer: 16.0 A)

Field of a solenoid: B = mu-zero (N / L) I.

So I = B L / (mu-zero N).

I = (0.4 x 0.5) / ((4 pi x 10^-7) x 10000).

I = 0.20 / 0.01256 = 15.9 A.

Answer: 16.0 A, as printed in the book.

Problem 14.8 - A galvanometer having an internal resistance Rg = 15.0 ohm gives full scale deflection with current Ig = 20.0 mA. It is to be converted into an ammeter of range 10.0 A. Find the value of shunt resistance Rs. (Book answer: 0.030 ohm)

The shunt is connected in parallel with the galvanometer, so the voltage across both is the same.

Ig Rg = (I - Ig) Rs, so Rs = Ig Rg / (I - Ig).

Ig = 20.0 mA = 0.020 A, Rg = 15.0 ohm, I = 10.0 A.

Rs = (0.020 x 15.0) / (10.0 - 0.020) = 0.30 / 9.98.

Answer: 0.030 ohm, as printed in the book.

Problem 14.9 - The resistance of a galvanometer is 50.0 ohm and reads full scale deflection with a current of 2.0 mA. Show by a diagram how to convert this galvanometer into voltmeter reading 200 V full scale. (Book answer: Rh = 99950 ohm)

To convert a galvanometer into a voltmeter, a high resistance Rh is connected in SERIES with it, as in the diagram: galvanometer - Rh - in series across the points whose potential difference is measured.

At full scale the current through the series combination is Ig = 2.0 mA = 0.002 A and the voltage across it is V = 200 V.

Total resistance needed: R = V / Ig = 200 / 0.002 = 100000 ohm.

So Rh = R - Rg = 100000 - 50.

Answer: Rh = 99950 ohm, as printed in the book.

Problem 14.10 - The resistance of a galvanometer coil is 10.0 ohm and reads full scale with a current of 1.0 mA. What should be the values of resistances R1, R2 and R3 to convert this galvanometer into a multirange ammeter of 100, 10.0 and 1.0 A as shown in the Fig. P. 14.10? (Book answers: R1 = 0.0001 ohm, R2 = 0.001 ohm, R3 = 0.01 ohm)

In Fig. P. 14.10 the galvanometer (Rg = 10.0 ohm, Ig = 1.0 mA = 0.001 A) is fitted with a shunt for each range, with taps at 100 A, 10 A and 1 A.

For a range I, the shunt Rs satisfies Ig Rg = (I - Ig) Rs, so Rs = Ig Rg / (I - Ig).

Ig Rg = 0.001 x 10.0 = 0.01 V.

100 A range: R1 = 0.01 / (100 - 0.001) = 0.0001 ohm.

10 A range: R2 = 0.01 / (10.0 - 0.001) = 0.001 ohm.

1.0 A range: R3 = 0.01 / (1.0 - 0.001) = 0.01 ohm.

Answers: R1 = 0.0001 ohm, R2 = 0.001 ohm, R3 = 0.01 ohm, as printed in the book.

Exercise - QUESTIONS 15.1 to 15.18 (book pp. 107-109, verbatim)

15.1 Does the induced emf in a circuit depend on the resistance of the circuit? Does the induced current depend on the resistance of the circuit?

The induced emf does NOT depend on the resistance of the circuit.

By Faraday's law, emf = -N (change in flux) / (change in time); it depends only on how fast the magnetic flux changes.

The induced current DOES depend on the resistance.

By Ohm's law, induced current = induced emf / resistance, so a larger resistance gives a smaller induced current for the same emf.

15.2 A square loop of wire is moving through a uniform magnetic field. The normal to the loop is oriented parallel to the magnetic field. Is a emf induced in the loop? Give a reason for your answer.

No, no emf is induced in the loop.

The flux through the loop is phi = B A, since the normal is parallel to the field and the field is uniform.

As the loop moves wholly inside the uniform field, neither B, nor the area, nor the orientation changes.

So the flux stays constant, its rate of change is zero, and by Faraday's law no emf is induced.

15.3 A light metallic ring is released from above into a vertical bar magnet (Fig. Q. 15.3). Viewed for above, does the current flow clockwise or anticlockwise in the ring?

Viewed from above, the current flows CLOCKWISE in the ring.

The ring falls towards the north pole of the magnet, so the upward magnetic flux through the ring increases as it approaches.

By Lenz's law, the induced current opposes this increase by creating a magnetic field pointing downward through the ring.

A clockwise current, viewed from above, produces a field pointing downward through the ring.

So the induced current is clockwise when viewed from above.

15.4 What is the direction of the current through resistor R in Fig. Q. 15.4? When switch S is (a) closed (b) opened. (Fig: primary coil with battery and switch S on the left, secondary coil with resistor R on the right, both coils wound the same way.)

When the switch is closed, current builds up in the primary and its magnetic field through the secondary grows, pointing to the right.

By Lenz's law the secondary current opposes this growth by making a field pointing to the left.

With the winding as drawn, that secondary current flows through resistor R from LEFT to RIGHT.

(a) Switch closed: current through R flows from left to right, while the primary field is building up.

When the switch is opened, the primary field to the right collapses.

The secondary current now opposes the decrease by making a field pointing to the right, the same way as the primary field did.

(b) Switch opened: current through R flows from right to left.

15.5 Does the induced emf always act to decrease the magnetic flux through a circuit?

No, the induced emf does not always act to decrease the flux.

Lenz's law says the induced emf opposes the CHANGE in flux, not the flux itself.

If the flux is increasing, the induced emf acts to decrease it.

But if the flux is decreasing, the induced emf acts to increase it, that is, to sustain it.

So the induced emf always fights the change, whichever way the flux is changing.

15.6 When the switch in the circuit is closed a current is established in the coil and the metal ring jumps upward (Fig. Q. 15.6). Why? Describe what would happen to the ring if the battery polarity were reversed?

Closing the switch makes the current, and hence the magnetic flux through the ring, grow rapidly.

By Lenz's law the ring carries an induced current whose magnetic field opposes the growth of the coil's field.

The ring's field and the coil's field repel each other, so the ring is pushed upward and jumps.

If the battery polarity were reversed, the coil's field would point the other way, but it would still be growing when the switch is closed.

The induced current would again oppose the growth, so the ring would STILL jump upward.

Reversing the polarity does not stop the jump; repulsion happens for either polarity.

15.7 The Fig. Q. 15.7 shows a coil of wire in the xy plane with a magnetic field directed along the y-axis. Around which of the three coordinate axes should the coil be rotated in order to generate an emf and a current in the coil?

The coil should be rotated about the x-axis.

The coil lies in the xy plane, so its normal is along the z-axis, perpendicular to the field along y, and the initial flux is zero.

Rotating about the x-axis tips the normal towards the y-axis, so the flux through the coil changes from zero to a growing value.

A changing flux induces an emf and a current.

Rotating about the y-axis keeps the normal perpendicular to the field, and rotating about the z-axis spins the coil in its own plane; in both cases the flux stays zero and no emf is generated.

15.8 How would you position a flat loop of wire in a changing magnetic field so that

there is no emf induced in the loop?

Position the loop so that its plane is parallel to the magnetic field.

Then the normal to the loop is perpendicular to the field, so cos theta = 0 and the flux through the loop is zero.

Even though the field strength is changing, the flux stays zero at all times.

With no change in flux, Faraday's law gives no induced emf.

15.9 In a certain region the earth's magnetic field point vertically down. When a plane flies due north, which wingtip is positively charged?

The west wingtip, the left wingtip as seen by a pilot facing north, becomes positively charged.

The free charges in the wing move north with the plane through a field pointing down.

The magnetic force q v cross B pushes positive charges towards the west and electrons towards the east.

So positive charge collects on the west (left) wingtip and the east (right) wingtip becomes negatively charged.

15.10 Show that epsilon and (change in flux) / (change in time) have the same units.

Faraday's law says emf = -N (change in flux) / (change in time), so the two sides must have the same units.

The unit of magnetic flux is the weber: 1 Wb = 1 T m squared.

Since 1 T = 1 N / (A m), we get 1 Wb = 1 N m / A = 1 J / A.

Dividing by time: 1 Wb / s = 1 J / (A s) = 1 J / C.

But 1 volt = 1 J / C, so 1 Wb / s = 1 V.

Hence (change in flux) / (change in time) has the unit volt, the same as emf.

15.11 When an electric motor, such as an electric drill, is being used, does it also act as a generator? If so what is the consequence of this?

Yes, a running motor also acts as a generator.

Its coil rotates in a magnetic field, so by electromagnetic induction an emf, the back emf, is induced in the coil.

The back emf opposes the supply voltage, so the net voltage driving current is supply voltage minus back emf.

As the motor speeds up, the back emf grows and the current drawn falls.

This is why a motor draws a large current at start-up, when it is not yet turning, and a much smaller current at full speed.

15.12 Can a D.C motor be turned into a D.C generator? What changes are required to be done?

Yes, a D.C. motor and a D.C. generator have the same construction; either can do the other's job.

To turn the motor into a generator, remove the supply battery and connect a load across its terminals instead.

Drive the shaft mechanically with an external prime mover, such as an engine or a turbine, instead of feeding it current.

The rotating coil then cuts the magnetic field and generates an emf, and the split-ring commutator delivers it as direct current.

No change to the coil, magnet or commutator is needed.

15.13 Is it possible to change both the area of the loop and the magnetic field passing through the loop and still not have an induced emf in the loop?

Yes, it is possible, provided the magnetic flux stays constant.

Flux is phi = B A cos theta, so if B increases while A decreases in the same proportion, the product B A stays the same.

With the flux unchanged, its rate of change is zero.

So by Faraday's law no emf is induced, even though both B and A are changing.

15.14 Can an electric motor be used to drive an electric generator with the output

from the generator being used to operate the motor?

No, such a self-running combination cannot work.

The generator's output is always less than the energy fed into it, because of friction, resistance heating and other losses.

If the generator fed the motor, the motor would get back less energy than it spent driving the generator.

The system would slow down and stop; it cannot sustain itself.

This would be a perpetual-motion machine, which the law of conservation of energy forbids.

15.15 A suspended magnet is oscillating freely in a horizontal plane. The oscillations are strongly damped when a metal plate is placed under the magnet. Explain why this occurs?

The oscillating magnet's changing magnetic flux through the metal plate induces eddy currents in the plate.

By Lenz's law, these eddy currents create magnetic fields that oppose the magnet's motion.

The opposing force drains the magnet's energy, which is dissipated as heat in the plate.

So the oscillations die away quickly: this is electromagnetic damping.

15.16 Four unmarked wires emerge from a transformer. What steps would you take to determine the turns ratio?

Step 1: Identify the two windings. Use a continuity tester or ohmmeter: the two wires of one winding show continuity with each other and a low resistance, while wires of different windings show no continuity.

Step 2: Label one winding as primary and the other as secondary.

Step 3: Apply a known alternating voltage to the primary winding and measure the voltage across the secondary winding with a voltmeter.

Step 4: The turns ratio is Ns / Np = Vs / Vp, the ratio of the measured secondary voltage to the applied primary voltage.

The winding with the higher voltage has the larger number of turns.

15.17 a) Can a step-up transformer increase the power level? b) In a transformer, there is no transfer of charge from the primary to the secondary. How is, then the power transferred?

(a) No, a step-up transformer cannot increase the power level.

For an ideal transformer, power out equals power in: Vs Is = Vp Ip.

Raising the voltage lowers the current in the same proportion, so the power stays the same, and in a real transformer it is slightly less because of losses.

(b) Power is transferred by the changing magnetic flux in the core, not by charge crossing over.

The alternating primary current creates a changing flux in the iron core.

This changing flux links the secondary coil and induces an emf in it, delivering power to the secondary circuit.

So energy flows through the magnetic field of the core, with no charge passing from primary to secondary.

15.18 When the primary of a transformer is connected to a.c mains the current in it a) is very small if the secondary circuit is open, but b) increases when the secondary circuit is closed. Explain these facts.

(a) Secondary open: no secondary current flows, so there is no secondary flux to oppose the primary flux.

The primary then needs only a small magnetizing current to set up the core flux, plus a little for core losses, so the primary current is very small.

(b) Secondary closed: the secondary current creates a flux that opposes the primary flux, by Lenz's law.

To keep the core flux at the value set by the mains voltage, the primary must draw

extra current to cancel this opposing flux.

So the primary current increases, roughly in proportion to the load on the secondary.

Exercise - PROBLEMS 15.1 to 15.18 (book pp. 109-110, verbatim, with the book's printed answers)

Problem 15.1 - An emf of 0.45 V is induced between the ends of a metal bar moving through a magnetic field of 0.22 T. What field strength would be needed to produce an emf of 1.5 V between the ends of the bar, assuming that all other factors remain the same? (Book answer: 0.73 T)

Motional emf is proportional to the field: emf is proportional to B, all else unchanged.

So B2 / B1 = emf2 / emf1.

B2 = 0.22 x (1.5 / 0.45).

B2 = 0.22 x 3.333 = 0.733.

Answer: 0.73 T, as printed in the book.

Problem 15.2 - The flux density B in a region between the pole faces of a horse-shoe magnet is 0.5 Wb m^-2 directed vertically downward. Find the emf induced in a straight wire 5.0 cm long, perpendicular to B when it is moved in a direction at an angle of 60 degrees with the horizontal with a speed of 100 cm s^-1. (Book answer: 1.25 x 10^-2 V)

Motional emf: emf = B L v-perpendicular, where v-perpendicular is the velocity

component perpendicular to the field.

The wire moves at 60 degrees to the horizontal, and the field is vertical, so the horizontal velocity component is perpendicular to the field.

Horizontal component: v cos 60 degrees = 100 cm s^-1 x 0.5 = 50 cm s^-1 = 0.50 m s^-1.

Length: L = 5.0 cm = 0.050 m, B = 0.5 T.

emf = 0.5 x 0.050 x 0.50 = 0.0125 V.

Answer: 1.25 x 10^-2 V, as printed in the book.

Problem 15.3 - A coil of wire has 10 loops. Each loop has an area of 1.5 x 10^-3 m squared. A magnetic field is perpendicular to the surface of each loop at all times. If the magnetic field is changed from 0.05 T to 0.06 T in 0.1 s, find the average emf induced in the coil during this time. (Book answer: +1.5 x 10^-3 V)

Faraday's law: emf = N A (change in B) / (change in time).

Change in B = 0.06 - 0.05 = 0.01 T. Time = 0.1 s. N = 10, A = 1.5 x 10^-3 m squared.

emf = 10 x (1.5 x 10^-3) x 0.01 / 0.1.

emf = 10 x (1.5 x 10^-3) x 0.1 = 1.5 x 10^-3 V.

Answer: +1.5 x 10^-3 V, as printed in the book (the positive sign follows the book's sign convention).

Problem 15.4 - A circular coil has 15 turns of radius 2 cm each. The plane of the coil lies at 40 degrees to a uniform magnetic field of 0.2 T. If the field is increased by 0.5 T in 0.2 s, find the magnitude of the induced emf. (Book answer: 1.8 x 10^-2 V - see note)

Step 1: Area of one turn: A = pi r squared = pi x (0.02) squared = 1.26 x 10^-3 m squared.

Step 2: The plane is at 40 degrees to the field, so the normal to the coil is at 50 degrees to the field, and cos 50 degrees = 0.643.

Step 3: Rate of change of field from the stem as printed: change in B / change in time = 0.5 / 0.2 = 2.5 T s^-1.

Step 4: emf = N A cos 50 degrees x (change in B / change in time).

emf = 15 x (1.26 x 10^-3) x 0.643 x 2.5 = 3.0 x 10^-2 V.

NOTE ON THE PRINTED ANSWER: the book prints 1.8 x 10^-2 V, which is the value obtained if the field change is 0.3 T, that is, the field rising FROM 0.2 T TO 0.5 T (0.5 - 0.2 = 0.3): 15 x (1.26 x 10^-3) x 0.643 x (0.3 / 0.2) = 1.8 x 10^-2 V. The step-by-step value from the stem as printed ('increased by 0.5 T') is 3.0 x 10^-2 V; the book's printed 1.8 x 10^-2 V is recorded here as printed, not silently substituted.

Problem 15.5 - Two coils are placed side by side. An emf of 0.8 V is observed in one coil when the current is changing at the rate of 200 A s^-1 in the other coil. What is the mutual inductance of the coils? (Book answer: 4 mH)

Mutual induction: emf = M x (rate of change of current).

So M = emf / (rate of change of current).

M = 0.8 / 200 = 0.004 H.

Answer: 4 mH, as printed in the book.

Problem 15.6 - A pair of adjacent coils has a mutual inductance of 0.75 H. If the current in the primary changes from 0 to 10 A in 0.025 s, what is the average induced emf in the secondary? What is the change in flux in it if the secondary has 500 turns? (Book answers: 300 V, 1.5 x 10^-2 Wb)

Step 1: Average induced emf: emf = M (change in I) / (change in time).

emf = 0.75 x (10 - 0) / 0.025 = 0.75 x 400 = 300 V.

Step 2: For the secondary, emf = N (change in flux) / (change in time), so change in flux = emf x (change in time) / N.

Change in flux = 300 x 0.025 / 500 = 7.5 / 500 = 0.015 Wb.

Answers: 300 V and 1.5 x 10^-2 Wb, as printed in the book.

Problem 15.7 - A solenoid has 250 turns and its self inductance is 2.4 mH. What is

the flux through each turn when the current is 2 A? What is the induced emf when the current changes at 20 A s^-1? (Book answers: 1.92 x 10^-5 Wb, 48 mV)

Step 1: Self inductance: L = N flux / I, so flux = L I / N.

Flux = (2.4 x 10^-3) x 2 / 250 = 4.8 x 10^-3 / 250.

Flux = 1.92 x 10^-5 Wb.

Step 2: Induced emf: emf = L (change in I) / (change in time).

emf = (2.4 x 10^-3) x 20 = 0.048 V.

Answers: 1.92 x 10^-5 Wb and 48 mV, as printed in the book.

Problem 15.8 - A solenoid of length 8.0 cm and cross sectional area 0.5 cm squared has 520 turns. Find the self inductance of the solenoid when the core is air. If the current in the solenoid increases through 1.5 A in 0.2 s, find the magnitude of induced emf in it. (mu-zero = 4 pi x 10^-7 Wb A^-1 m^-1) (Book answers: 1.6 x 10^-3 V, 2.12 x 10^-4 H)

Step 1: Self inductance of a solenoid: L = mu-zero N squared A / l.

N squared = 520 squared = 270400. A = 0.5 cm squared = 0.5 x 10^-4 m squared. l = 8.0 cm = 0.080 m.

L = (4 pi x 10^-7) x 270400 x (0.5 x 10^-4) / 0.080.

L = 2.12 x 10^-4 H.

Step 2: Induced emf: emf = L (change in I) / (change in time).

emf = (2.12 x 10^-4) x 1.5 / 0.2 = (2.12 x 10^-4) x 7.5.

emf = 1.6 x 10^-3 V.

Answers: 1.6 x 10^-3 V and 2.12 x 10^-4 H, as printed in the book.

Problem 15.9 - When current through a coil changes from 100 mA to 200 mA in 0.005 s, an induced emf of 40 mV is produced in the coil. (a) What is the self inductance of the coil? (b) Find the increase in the energy stored in the coil. (Book answers: 2 mH, 0.03 mJ)

Step 1: (a) emf = L (change in I) / (change in time), so L = emf x (change in time) / (change in I).

Change in I = 200 - 100 = 100 mA = 0.100 A. emf = 40 mV = 0.040 V.

L = 0.040 x 0.005 / 0.100 = 0.002 H.

Step 2: (b) Energy stored: U = (1/2) L I squared. Increase = (1/2) L (I2 squared - I1 squared).

Increase = 0.5 x 0.002 x ((0.200) squared - (0.100) squared).

Increase = 0.001 x (0.040 - 0.010) = 0.001 x 0.030 = 3.0 x 10^-5 J.

Answers: 2 mH and 0.03 mJ, as printed in the book.

Problem 15.10 - Like any field, the earth's magnetic field stores energy. Find the magnetic energy stored in a space where strength of earth's field is 7 x 10^-5 T, if the space occupies an area of 10 x 10^8 m squared and has a height of 750 m. (Book answer: 1.46 x 10^9 J)

Step 1: Energy density in a magnetic field: u = B squared / (2 mu-zero).

u = (7 x 10^-5) squared / (2 x 4 pi x 10^-7) = (49 x 10^-10) / (2.51 x 10^-6).

u = 1.95 x 10^-3 J m^-3.

Step 2: Volume: V = area x height = (10 x 10^8) x 750 = 7.5 x 10^11 m cubed.

Step 3: Total energy: U = u V = (1.95 x 10^-3) x (7.5 x 10^11).

U = 1.46 x 10^9 J.

Answer: 1.46 x 10^9 J, as printed in the book.

Problem 15.11 - A square coil of side 16 cm has 200 turns and rotates in a uniform magnetic field of magnitude 0.05 T. If the peak emf is 12 V, what is the angular velocity of the coil (Book answer: 47 rad s^-1)

Peak emf of a rotating coil: emf-zero = N A B omega.

Area: A = 0.16 x 0.16 = 0.0256 m squared.

So omega = emf-zero / (N A B) = 12 / (200 x 0.0256 x 0.05).

omega = 12 / 0.256 = 46.9 rad s^-1.

Answer: 47 rad s^-1, as printed in the book.

Problem 15.12 - A generator has a rectangular coil consisting of 360 turns. The coil rotates at 420 rev per min in 0.14 T magnetic field. The peak value of emf produced by the generator is 50 V. If the coil is 5.0 cm wide, find the length of the side of the coil. (Book answer: 45 cm)

Step 1: Frequency: f = 420 / 60 = 7 rev s^-1. Angular velocity: omega = 2 pi f = 2 pi x 7 = 44.0 rad s^-1.

Step 2: Peak emf: emf-zero = N A B omega, so A = emf-zero / (N B omega).

A = 50 / (360 x 0.14 x 44.0) = 50 / 2218 = 0.0225 m squared.

Step 3: Area = width x length, so length = A / width = 0.0225 / 0.050.

Length = 0.45 m.

Answer: 45 cm, as printed in the book.

Problem 15.13 - It is desired to make an a.c generator that can produce an emf of maximum value 5 kV with 50 Hz frequency. A coil of area 1 m squared having 200 turns is used as armature. What should be the magnitude of the magnetic field in which the coil rotates? (Book answer: 0.08 T)

Peak emf: emf-zero = N A B omega, with omega = 2 pi f.

omega = 2 pi x 50 = 314 rad s^-1.

So B = emf-zero / (N A omega) = 5000 / (200 x 1 x 314).

B = 5000 / 62800 = 0.0796 T.

Answer: 0.08 T, as printed in the book.

Problem 15.14 - The back emf in a motor is 120 V when the motor is turning at 1680 rev per min. What is the back emf when the motor turns 3360 rev per min? (Book answer: 240 V)

Back emf is proportional to the speed of rotation.

So emf2 / emf1 = speed2 / speed1 = 3360 / 1680 = 2.

emf2 = 120 x 2 = 240 V.

Answer: 240 V, as printed in the book.

Problem 15.15 - A D.C motor operates at 240 V and has a resistance of 0.5 ohm. When the motor is running at normal speed, the armature current is 15 A. Find the back emf in the armature. (Book answer: 232.5 V)

For a motor: supply voltage V = back emf + I R.

So back emf = V - I R.

Back emf = 240 - (15 x 0.5) = 240 - 7.5.

Answer: 232.5 V, as printed in the book.

Problem 15.16 - A copper ring has a radius of 4.0 cm and resistance of 1.0 m ohm. A magnetic field is applied over the ring, perpendicular to its plane. If the magnetic field increases from 0.2 T to 0.4 T in a time interval of 5 x 10^-3 s, what is the current in the ring during this interval? (Book answer: 201 A)

Step 1: Area of the ring: A = pi r squared = pi x (0.040) squared = 5.03 x 10^-3 m squared.

Step 2: Induced emf: emf = A (change in B) / (change in time).

emf = (5.03 x 10^-3) x (0.4 - 0.2) / (5 x 10^-3) = (5.03 x 10^-3) x 40.

emf = 0.201 V.

Step 3: Current: I = emf / R, with R = 1.0 m ohm = 1.0 x 10^-3 ohm.

I = 0.201 / (1.0 x 10^-3) = 201 A.

Answer: 201 A, as printed in the book.

Problem 15.17 - A coil of 10 turns and 35 cm squared area is in a perpendicular magnetic field of 0.5 T. The coil is pulled out of the field in 1.0 s. Find the induced emf in the coil as it is pulled out of the field. (Book answer: 1.75 x 10^-2 V)

Flux through the coil falls from B A to zero in 1.0 s.

emf = N A B / time.

A = 35 cm squared = 35 x 10^-4 m squared.

emf = 10 x (35 x 10^-4) x 0.5 / 1.0 = 10 x 0.0035 x 0.5.

Answer: 1.75 x 10^-2 V, as printed in the book.

Problem 15.18 - An ideal step down transformer is connected to main supply of 240 V. It is desired to operate a 12 V, 30 W lamp. Find the current in the primary and the transformation ratio? (Book answers: 0.125 A, 1/20)

Step 1: For an ideal transformer, power in = power out = 30 W.

Primary current: Ip = P / Vp = 30 / 240 = 0.125 A.

Step 2: Transformation ratio: Ns / Np = Vs / Vp = 12 / 240.

Ratio = 1 / 20.

Answers: 0.125 A and 1/20, as printed in the book. (Secondary current, for reference, is 30 / 12 = 2.5 A.)

Chapter 16: Alternating Current

Exercise - QUESTIONS 16.1 to 16.10 (book pp. 132-133, verbatim)

16.1 A sinusoidal current has rms value of 10 A. What is the maximum or peak value?

For a sinusoidal current, the rms value is the peak value divided by the square root of 2: I-rms = I-zero / sqrt(2).

So the peak value is I-zero = I-rms x sqrt(2).

I-zero = 10 x 1.414 = 14.14 A.

Answer: the maximum or peak value is about 14.1 A.

16.2 Name the device that will (a) permit flow of direct current but oppose the flow of alternating current (b) permit flow of alternating current but not the direct current.

(a) An inductor, or choke coil, permits direct current but opposes alternating current. Its inductive reactance is X-L = 2 pi f L. For direct current f = 0, so X-L = 0 and the current passes freely.

For alternating current f is not zero, so X-L opposes the current.

(b) A capacitor permits alternating current but not direct current.

Its capacitive reactance is X-C = 1 / (2 pi f C). For direct current f = 0, so X-C is infinite and no steady current passes.

For alternating current X-C is finite, so the current passes.

16.3 How many times per second will an incandescent lamp reach maximum brilliance when connected to a 50 Hz source?

The brilliance of the lamp follows the power, which follows the square of the current. In one cycle of a 50 Hz supply, the current reaches its positive peak once and its negative peak once.

The square of the current, and hence the brilliance, is a maximum at both peaks.

So the lamp reaches maximum brilliance twice per cycle.

Answer: 2 x 50 = 100 times per second.

16.4 A circuit contains an iron-cored inductor, a switch and a D.C. source arranged in series. The switch is closed and after an interval reopened. Explain why a spark jumps across the switch contacts?

While the switch is closed, a steady current flows and the inductor stores energy in its magnetic field.

When the switch is reopened, the current tries to fall to zero suddenly, so the rate of change of current is very large.

The inductor opposes this change by inducing a very large emf, emf = -L (change in current) / (change in time).

This large induced emf appears across the opening switch contacts.

It is high enough to break down the air gap, ionize it, and a spark jumps across the contacts.

16.5 How does doubling the frequency affect the reactance of (a) an inductor (b) a capacitor?

(a) Inductor: X-L = 2 pi f L, so reactance is directly proportional to frequency.

Doubling the frequency doubles the inductive reactance.

(b) Capacitor: X-C = 1 / (2 pi f C), so reactance is inversely proportional to frequency.

Doubling the frequency halves the capacitive reactance.

16.6 In a R - L circuit, will the current lag or lead the voltage? Illustrate your answer by a vector diagram.

In an R-L circuit the current LAGS the voltage.

The voltage across the resistor, V-R, is in phase with the current I.

The voltage across the inductor, V-L, leads the current by 90 degrees, because the inductor opposes changes in current.

Vector diagram: draw the current vector I as the reference along the horizontal.

Draw V-R along I, and V-L 90 degrees ahead of (above) I.

The total applied voltage V is the vector sum of V-R and V-L, and it lies ahead of I by the phase angle phi, where tan phi = X-L / R.

So the current lags behind the applied voltage by phi.

16.7 A choke coil placed in series with an electric lamp in an A.C. circuit causes the lamp to become dim. Why is it so? A variable capacitor added in series in this circuit may be adjusted until the lamp glows with normal brilliance. Explain, how this is possible?

The choke coil adds inductive reactance X-L to the circuit, so the total impedance Z becomes larger.

A larger impedance means a smaller current, so the lamp, which now gets less current, becomes dim.

Adding a variable capacitor in series adds capacitive reactance X-C, which subtracts from X-L in the impedance.

By adjusting the capacitance until X-C = X-L, the circuit reaches resonance.

At resonance the impedance is at its minimum, equal to the resistance alone, so the current is at its maximum.

The lamp then gets its full current again and glows with normal brilliance.

16.8 Explain the conditions under which electromagnetic waves are produced from a source?

Electromagnetic waves are produced by accelerating electric charges.

A charge oscillating back and forth, for example in a transmitting antenna, creates a changing electric field.

The changing electric field creates a changing magnetic field, which in turn creates a changing electric field, and so on.

These mutually sustaining changing fields detach from the source and travel outwards through space.

No medium is needed: the waves propagate even through vacuum at the speed of light.

16.9 How the reception of a particular radio station is selected on your radio set?

The radio set has a tuning circuit, a coil and a variable capacitor in parallel, with a

natural resonant frequency f = 1 / (2 pi sqrt(L C)).

By turning the tuning knob, the capacitance is changed until the circuit's resonant frequency equals the carrier frequency of the wanted station.

At resonance, that station's signal builds up a large response in the circuit, while other stations, off resonance, give only a tiny response.

So the wanted station is selected and the others are rejected.

16.10 What is meant by A.M. and F.M.?

A.M. means amplitude modulation. The amplitude of a high-frequency carrier wave is varied in step with the low-frequency signal, while the carrier frequency stays constant.

F.M. means frequency modulation. The frequency of the carrier wave is varied in step with the signal, while the carrier amplitude stays constant.

In both cases the signal is carried by the modulated carrier wave to the receiver, which recovers the original signal.

Exercise - PROBLEMS 16.1 to 16.10 (book pp. 133-134, verbatim, with the book's printed answers)

Problem 16.1 - An alternating current is represented by the equation I = 20 sin 100 pi t. Compute its frequency and the maximum and rms values of current. (Book answers: 50 Hz, 20 A, 14 A)

Compare with I = I-zero sin (omega t): omega = 100 pi.

Frequency: f = omega / (2 pi) = 100 pi / (2 pi) = 50 Hz.

Maximum value: I-zero = 20 A.

rms value: I-rms = I-zero / sqrt(2) = 20 / 1.414 = 14.14 A.

Answers: 50 Hz, 20 A, 14 A, as printed in the book.

Problem 16.2 - A sinusoidal A.C. has a maximum value of 15 A. What are its rms values? If the time is recorded from the instant the current is zero and is becoming positive, what is the instantaneous value of the current after 1 / 300 s, given the frequency is 50 Hz. (Book answers: I-rms = 10.6 A, Instantaneous current = 13.0 A)

Step 1: rms value: I-rms = I-zero / sqrt(2) = 15 / 1.414 = 10.6 A.

Step 2: Instantaneous current: I = I-zero sin (omega t), with omega = 2 pi f = 2 pi x 50 = 100 pi rad s^-1.

At t = 1 / 300 s: omega t = 100 pi / 300 = pi / 3 = 60 degrees.

I = 15 x sin 60 degrees = 15 x 0.866 = 13.0 A.

Answers: I-rms = 10.6 A and instantaneous current = 13.0 A, as printed in the book.

Problem 16.3 - Find the value of the current and inductive reactance when A.C. voltage of 220 V at 50 Hz is passed through an inductor of 10 H. (Book answers: I-rms = 0.07 A, X-L = 3140 ohm)

Inductive reactance: X-L = 2 pi f L.

X-L = 2 pi x 50 x 10 = 3140 ohm.

Current: I-rms = V-rms / X-L = 220 / 3140 = 0.070 A.

Answers: I-rms = 0.07 A and X-L = 3140 ohm, as printed in the book.

Problem 16.4 - A circuit has an inductance of 1 / pi H and resistance of 2000 ohm. A 50 Hz A.C. is supplied to it. Calculate the reactance and impedance offered by the circuit. (Book answers: X-L = 100 ohm, Z = 2002.5 ohm)

Inductive reactance: X-L = 2 pi f L = 2 pi x 50 x (1 / pi) = 100 ohm.

Impedance: Z = sqrt(R squared + X-L squared) = sqrt((2000) squared + (100) squared).

Z = sqrt(4000000 + 10000) = sqrt(4010000) = 2002.5 ohm.

Answers: X-L = 100 ohm and Z = 2002.5 ohm, as printed in the book.

Problem 16.5 - An inductor of pure inductance 3 / pi H is connected in series with a resistance of 40 ohm. Find (i) the peak value of the current (ii) the rms value, and (iii) the phase difference between the current and the applied voltage V = 350 sin (100 pi t). (Book answers: (i) 1.16 A, (ii) 0.81 A, (iii) 82.4 degrees)

Step 1: From V = 350 sin (100 pi t): V-zero = 350 V, omega = 100 pi rad s^-1.

Step 2: Inductive reactance: X-L = omega L = 100 pi x (3 / pi) = 300 ohm.

Step 3: Impedance: Z = sqrt(R squared + X-L squared) = sqrt((40) squared + (300) squared) = sqrt(91600) = 302.7 ohm.

Step 4: (i) Peak current: I-zero = V-zero / Z = 350 / 302.7 = 1.16 A.

Step 5: (ii) rms current: I-rms = I-zero / sqrt(2) = 1.16 / 1.414 = 0.82 A by unrounded division; the book prints 0.81 A.

Step 6: (iii) Phase difference: tan phi = X-L / R = 300 / 40 = 7.5, so phi = 82.4 degrees (current lags voltage).

Answers: (i) 1.16 A, (ii) 0.81 A as printed, (iii) 82.4 degrees, as printed in the book.

Problem 16.6 - A 10 mH, 20 ohm coil is connected across 240 V and 180 / pi Hz source. How much power does it dissipate? (Book answer: 2778 W)

Step 1: Angular frequency: omega = 2 pi f = 2 pi x (180 / pi) = 360 rad s^-1.

Step 2: Inductive reactance: X-L = omega L = 360 x (10 x 10^-3) = 3.6 ohm.

Step 3: Impedance: Z = sqrt(R squared + X-L squared) = sqrt((20) squared + (3.6) squared) = sqrt(412.96) = 20.32 ohm.

Step 4: rms current: I-rms = V / Z = 240 / 20.32 = 11.81 A.

Step 5: Power dissipated (in the resistance only): P = (I-rms) squared x R = (11.81) squared x 20 = 2790 W by unrounded computation; the book prints 2778 W, the small difference coming from intermediate rounding in the book's working.

Answer: 2778 W as printed in the book.

Problem 16.7 - Find the value of the current flowing through a capacitance 0.5 microF when connected to a source of 150 V at 50 Hz. (Book answer: I-rms = 0.024 A)

Capacitive reactance: X-C = 1 / (2 pi f C).

X-C = 1 / (2 pi x 50 x (0.5 x 10^-6)) = 1 / (1.571 x 10^-4) = 6366 ohm.

Current: I-rms = V / X-C = 150 / 6366 = 0.0236 A.

Answer: I-rms = 0.024 A, as printed in the book.

Problem 16.8 - An alternating source of emf 12 V and frequency 50 Hz is applied to a capacitor of capacitance 3 microF in series with a resistor of resistance 1k ohm. Calculate the phase angle. (Book answer: 46.7 degrees)

Capacitive reactance: X-C = 1 / (2 pi f C) = 1 / (2 pi x 50 x (3 x 10^-6)).

X-C = 1 / (9.425 x 10^-4) = 1061 ohm.

Resistance: R = 1k ohm = 1000 ohm.

Phase angle: tan phi = X-C / R = 1061 / 1000 = 1.061, so phi = 46.7 degrees.

Answer: 46.7 degrees, as printed in the book (current leads voltage in this R-C circuit).

Problem 16.9 - What is the resonant frequency of a circuit which includes a coil of inductance 2.5 H and a capacitance 40 microF? (Book answer: 15.9 Hz)

Resonant frequency: f-zero = 1 / (2 pi sqrt(L C)).

L C = 2.5 x (40 x 10^-6) = 1.0 x 10^-4.

sqrt(L C) = sqrt(1.0 x 10^-4) = 0.010.

f-zero = 1 / (2 pi x 0.010) = 1 / 0.0628 = 15.9 Hz.

Answer: 15.9 Hz, as printed in the book.

Problem 16.10 - An inductor of inductance 150 microH is connected in parallel with a variable capacitor whose capacitance can be changed from 500 pF to 20 pF. Calculate the maximum frequency and minimum frequency for which the circuit can be tuned. (Book answers: 2.91 MHz, 0.58 MHz)

Resonant frequency: f-zero = 1 / (2 pi sqrt(L C)). The frequency is highest for the smallest capacitance and lowest for the largest.

Maximum frequency (C = 20 pF): L C = (150 x 10^-6) x (20 x 10^-12) = 3.0 x 10^-15.

f-max = 1 / (2 pi x sqrt(3.0 x 10^-15)) = 1 / (2 pi x 5.48 x 10^-8) = 2.91 x 10^6 Hz.

Minimum frequency (C = 500 pF): L C = (150 x 10^-6) x (500 x 10^-12) = 7.5 x 10^-14.

f-min = 1 / (2 pi x sqrt(7.5 x 10^-14)) = 1 / (2 pi x 2.74 x 10^-7) = 0.58 x 10^6 Hz.

Answers: 2.91 MHz and 0.58 MHz, as printed in the book.

Chapter 17: Physics of Solids

Exercise - QUESTIONS 17.1 to 17.11 (book p. 152, verbatim)

17.1 Distinguish between crystalline, amorphous and polymeric solids.

Crystalline solids: atoms arranged in a regular, repeating three-dimensional pattern, a lattice.

They have sharp melting points and their properties may differ in different directions.

Examples: common salt, diamond, metals.

Amorphous solids: atoms have no long-range regular arrangement; they are disordered like a frozen liquid.

They soften gradually over a range of temperature instead of melting sharply.

Examples: glass, rubber, plastics.

Polymeric solids: made of very long chain molecules of repeating units.

They are partly ordered and partly disordered, light, and often flexible.

Examples: polyethylene, nylon, PVC.

17.2 Define stress and strain. What are their SI units? Differentiate between tensile, compressive and shear modes of stress and strain.

Stress is the deforming force per unit area: stress = F / A.

Its SI unit is the newton per square metre, N m^-2, called the pascal (Pa).

Strain is the fractional change in dimension: strain = change in dimension / original dimension.

It is a pure ratio, so it has no unit.

Tensile mode: a pulling force stretches the body; tensile stress pulls outward on the area and tensile strain is the fractional increase in length.

Compressive mode: a pushing force squeezes the body; compressive stress pushes inward and compressive strain is the fractional decrease in length.

Shear mode: a tangential force acts parallel to a face; shear stress is the tangential force per unit area and shear strain is the angular distortion produced.

17.3 Define modulus of elasticity. Show that the units of modulus of elasticity and stress are the same. Also discuss its three kinds.

Modulus of elasticity is the ratio of stress to strain, within the elastic limit: modulus = stress / strain.

Stress has the unit pascal, and strain has no unit, being a pure ratio.

So modulus = pascal / (no unit) = pascal, the same unit as stress.

Its three kinds are: Young's modulus, for longitudinal stress and strain, Y = (F / A) / (change in length / length).

Bulk modulus, for volume stress and strain, K = (change in pressure) / (fractional change in volume).

Shear modulus (modulus of rigidity), for shear stress and strain, G = (tangential stress) / (shear strain).

17.4 Draw a stress-strain curve for a ductile material, and then define the terms: Elastic limit, Yield point and Ultimate tensile stress.

Stress-strain curve for a ductile material (described): stress rises linearly with strain at first (proportional region), then bends over at the elastic limit.

After the yield point the curve flattens: strain grows a lot for little extra stress.

It then rises again (strain hardening) to a maximum, the ultimate tensile stress, falls during necking, and ends at the fracture point.

Elastic limit: the greatest stress the material can take and still return fully to its original shape when the stress is removed.

Yield point: the stress at which the material begins to deform plastically, that is, strain increases rapidly with little or no increase in stress.

Ultimate tensile stress: the maximum stress the material can withstand before necking and fracture; the highest point of the stress-strain curve.

17.5 What is meant by strain energy? How can it be determined from the force-extension graph?

Strain energy is the energy stored in a body because it has been deformed within its elastic limit.

It is the work done in deforming the body, and it is recovered when the deforming force is removed.

From the force-extension graph, the strain energy is the area under the graph up to the extension reached.

For a material obeying Hooke's law the graph is a straight line, so the area is a triangle: strain energy = (1/2) x force x extension.

In general, strain energy per unit volume = (1/2) x stress x strain.

17.6 Describe the formation of energy bands in solids. Explain the difference amongst electrical behaviour of conductors, insulators and semi-conductors in terms of energy band theory.

In an isolated atom, electrons occupy sharp, discrete energy levels.

In a solid, huge numbers of atoms are packed close together, and their outer energy levels interact and split into very many closely spaced levels.

These merge into continuous ranges called energy bands, separated by forbidden gaps where no electron states exist.

The lower filled band is the valence band and the upper band is the conduction band.

Conductors: the valence and conduction bands overlap, or the conduction band is partly filled, so electrons move freely and conduct easily.

Insulators: the forbidden gap is very large (about 5 eV or more), so almost no electron can jump to the conduction band, and conduction is negligible.

Semi-conductors: the forbidden gap is small (about 1 eV), so at room temperature some electrons cross into the conduction band and conduct, and conductivity rises with temperature.

17.7 Distinguish between intrinsic and extrinsic semi-conductors. How would you

obtain n-type and p-type material from pure silicon? Illustrate it by schematic diagram.

Intrinsic semi-conductors are pure, with no added impurity; their charge carriers are equal numbers of electrons and holes produced by thermal breaking of bonds.

Extrinsic semi-conductors are doped, that is, a small number of impurity atoms are added to pure silicon to increase one kind of carrier.

n-type from pure silicon: dope with a pentavalent impurity such as phosphorus.

Four of its valence electrons form covalent bonds with neighbouring silicon atoms, and the fifth electron is free to move in the crystal.

Schematic diagram: a silicon lattice in which one silicon atom is replaced by a phosphorus atom with four bonds and one free electron shown beside it.

p-type from pure silicon: dope with a trivalent impurity such as boron.

It forms only three bonds, leaving a vacancy, a hole, which acts as a positive charge carrier.

Schematic diagram: a silicon lattice in which one silicon atom is replaced by a boron atom with three bonds and one hole shown beside it.

17.8 Discuss the mechanism of electrical conduction by holes and electrons in a pure semi-conductor element.

In a pure semi-conductor, heat energy breaks some covalent bonds, freeing an electron and leaving a hole behind.

The free electron moves in the conduction band towards the positive terminal.

At the same time, a neighbouring bond electron can jump into the hole, so the hole moves in the opposite direction, towards the negative terminal.

The hole thus behaves like a positive charge carrier moving through the valence band.

Both the electrons and the holes carry current, so the total current is the sum of the electron current and the hole current.

17.9 Write a note on superconductors.

Superconductors are materials whose electrical resistivity falls suddenly to zero below a certain critical temperature, Tc.

Below Tc they conduct current with no resistance and no energy loss, so a current once started can flow indefinitely.

They also show perfect diamagnetism: they expel magnetic fields from inside them, the Meissner effect.

Examples: mercury becomes superconducting below 4.2 K; niobium alloys and ceramic oxides are used at higher temperatures.

Uses: powerful electromagnets for MRI scanners and particle accelerators, and loss-free power transmission and magnetic levitation.

17.10 What is meant by para, dia and ferromagnetic substances? Give examples for each.

Paramagnetic substances: weakly attracted by a magnetic field, with a small positive susceptibility.

Their atoms have unpaired electrons whose magnetic moments partly align with an applied field.

Examples: aluminium, platinum, oxygen.

Diamagnetic substances: weakly repelled by a magnetic field, with a small negative susceptibility.

Their atoms have paired electrons, so the induced moments oppose the applied field.

Examples: copper, bismuth, water.

Ferromagnetic substances: strongly attracted by a magnetic field, with a very large positive susceptibility.

Their atomic moments form domains that align strongly with the field, and they can stay magnetized.

Examples: iron, nickel, cobalt.

17.11 What is meant by hysteresis loss? How is it used in the construction of a transformer?

Hysteresis loss is the energy lost as heat in a magnetic material in each cycle of magnetization and demagnetization.

It equals the area of the hysteresis loop for that material, and it is lost in every alternating-current cycle.

In a transformer the core is magnetized and demagnetized 50 times per second, so a large loop would waste a lot of energy.

Therefore transformer cores are made of soft magnetic material with a narrow hysteresis loop, such as soft iron or silicon steel.

A narrow loop means a small area and small hysteresis loss; the core is also laminated to reduce eddy-current loss.

Exercise - PROBLEMS 17.1 to 17.6 (book p. 153, verbatim, with the book's printed answers)

Problem 17.1 - A 1.25 cm diameter cylinder is subjected to a load of 2500 kg. Calculate the stress on the bar in mega pascals. (Book answer: 200 MPa)

Force: F = mass x g = 2500 x 9.8 = 24500 N.

Radius: r = diameter / 2 = 1.25 / 2 = 0.625 cm = 6.25 x 10^-3 m.

Cross-sectional area: A = pi r squared = pi x (6.25 x 10^-3) squared = 1.227 x 10^-4 m squared.

Stress = F / A = 24500 / (1.227 x 10^-4) = 1.997 x 10^8 Pa.

Answer: 200 MPa, as printed in the book.

Problem 17.2 - A 1.0 m long copper wire is subjected to stretching force and its length increases by 20 cm. Calculate the tensile strain and the percent elongation which the wire undergoes. (Book answers: 0.20, 20%)

Tensile strain = change in length / original length.

Change in length = 20 cm = 0.20 m, original length = 1.0 m.

Strain = 0.20 / 1.0 = 0.20.

Percent elongation = strain x 100 = 20%.

Answers: 0.20 and 20%, as printed in the book.

Problem 17.3 - A wire 2.5 m long and cross-section area 10^-5 m squared is stretched 1.5 mm by a force of 100 N in the elastic region. Calculate (i) the strain (ii) Young's modulus (iii) the energy stored in the wire. (Book answers: 6.02 x 10^-4, 1.66 x 10^10 Pa, 7.5 x 10^-2 J)

Step 1: (i) Strain = change in length / original length = (1.5 x 10^-3) / 2.5 = 6.0 x 10^-4.

Step 2: Stress = F / A = 100 / (10^-5) = 1.0 x 10^7 Pa.

Step 3: (ii) Young's modulus Y = stress / strain = (1.0 x 10^7) / (6.0 x 10^-4) = 1.67 x 10^10 Pa.

Step 4: (iii) Energy stored = (1/2) x force x extension = 0.5 x 100 x (1.5 x 10^-3) = 7.5 x 10^-2 J.

NOTE: the book prints the pair (6.02 x 10^-4, 1.66 x 10^10 Pa), which is internally consistent (1.0 x 10^7 / 6.02 x 10^-4 = 1.66 x 10^10) but differs slightly from the exact division above; the energy answer agrees either way. The book's printed values are recorded here as printed.

Problem 17.4 - What stress would cause a wire to increase in length by 0.01% if the Young's modulus of the wire is 12 x 10^10 Pa. What force would produce this stress if the diameter of the wire is 0.56 mm? (Book answers: 1.2 x 10^9 Pa, 2.96 N)

Step 1: Strain = 0.01% = 0.01 / 100 = 1.0 x 10^-4.

Step 2: Stress = Y x strain = (12 x 10^10) x (1.0 x 10^-4) = 1.2 x 10^7 Pa.

Step 3: Area: radius = 0.56 / 2 = 0.28 mm = 2.8 x 10^-4 m. A = pi r squared = pi x (2.8 x 10^-4) squared = 2.46 x 10^-7 m squared.

Step 4: Force = stress x area = (1.2 x 10^7) x (2.46 x 10^-7) = 2.96 N.

NOTE ON THE PRINTED ANSWER: the book prints the stress as 1.2 x 10^9 Pa, but that exponent cannot be right, because with 1.2 x 10^9 Pa the force would be 296 N, not 2.96 N. The printed force 2.96 N itself requires the stress 1.2 x 10^7 Pa obtained above, so the book's stress exponent is a misprint. Answers: stress 1.2 x 10^7 Pa (book prints 1.2 x 10^9 Pa) and force 2.96 N, as printed in the book.

Problem 17.5 - The length of a steel wire is 1.0 m and its cross-sectional area is 0.03

x 10^-4 m squared. Calculate the work done in stretching the wire when a force of 100 N is applied within the elastic region. Young's modulus of steel is 3.0 x 10^11 N m^-2. (Book answer: 5.6 x 10^-3 J)

Step 1: Extension: change in length = F L / (A Y).

Change in length = (100 x 1.0) / ((0.03 x 10^-4) x (3.0 x 10^11)).

Change in length = 100 / (9.0 x 10^5) = 1.11 x 10^-4 m.

Step 2: Work done = (1/2) x force x extension = 0.5 x 100 x (1.11 x 10^-4).

Work done = 5.6 x 10^-3 J.

Answer: 5.6 x 10^-3 J, as printed in the book.

Problem 17.6 - A cylindrical copper wire and a cylindrical steel wire each of length 1.5 m and diameter 2.0 mm are joined at one end to form a composite wire 3.0 m long. The wire is loaded until its length becomes 3.003 m. Calculate the strain in copper and steel wires and the force applied to the wire. (Young's modulus of copper is 1.2 x 10^11 Pa and for steel is 2.0 x 10^11 Pa). (Book answers: 1.25 x 10^-3, 7.5 x 10^-4, 477 N)

Step 1: Both wires carry the same force F (they are in series), and their extensions add to 3.003 - 3.0 = 0.003 m.

Step 2: Area of each wire: radius = 1.0 mm = 1.0 x 10^-3 m, A = pi r squared = 3.14 x 10^-6 m squared.

Step 3: Extension of each wire = F L / (A Y). With L = 1.5 m for each: total extension = (F x 1.5 / A) x (1 / Y-copper + 1 / Y-steel).

So 0.003 = (F x 1.5 / (3.14 x 10^-6)) x (1 / (1.2 x 10^11) + 1 / (2.0 x 10^11)) = F x (1.5 x 1.333 x 10^-11) / (3.14 x 10^-6) = F x 6.37 x 10^-6.

Hence F = 0.003 / (6.37 x 10^-6) = 471 N.

Step 4: Strain in copper = F / (A Y-copper) = 471 / ((3.14 x 10^-6) x (1.2 x 10^11)) = 1.25 x 10^-3.

Step 5: Strain in steel = F / (A Y-steel) = 471 / ((3.14 x 10^-6) x (2.0 x 10^11)) = 7.5 x 10^-4.

Check: extensions are 1.25 x 10^-3 x 1.5 = 1.875 x 10^-3 m and 7.5 x 10^-4 x 1.5 = 1.125 x 10^-3 m, totalling 0.003 m, as required.

NOTE: the book prints the force as 477 N, while the printed strains give 471 N by the steps above; the strains and the total extension agree exactly. The book's printed 477 N is recorded here as printed.

Chapter 18: Electronics

Exercise - QUESTIONS 18.1 to 18.11 (book p. 172, verbatim)

18.1 How does the motion of an electron in a n-type substance differ from the

motion of holes in a p-type substance?

In an n-type substance the current is carried mainly by free electrons in the conduction band.

These electrons move opposite to the applied electric field, towards the positive terminal.

In a p-type substance the current is carried mainly by holes in the valence band.

A hole moves when a neighbouring bond electron jumps into it, so the hole drifts in the direction of the field, opposite to the electron motion.

So electrons and holes move in opposite directions under the same field, and the n-type majority carriers are electrons while the p-type majority carriers are holes.

18.2 What is the net charge on a n-type or a p-type substance?

The net charge on both is zero; both are electrically neutral.

Doping adds neutral impurity atoms to neutral silicon, so no net charge is added.

In n-type material the extra free electrons are balanced by positively charged donor ions fixed in the lattice.

In p-type material the holes are balanced by negatively charged acceptor ions fixed in the lattice.

So although one carrier type dominates the conduction, the material as a whole stays neutral.

18.3 The anode of a diode is 0.2 V positive with respect to its cathode. Is it forward biased?

Yes, it is forward biased, because the anode (p-side) is positive with respect to the cathode (n-side).

Forward bias means the p-side is made positive, which narrows the depletion region and allows current to flow.

However, 0.2 V is below the knee voltage, about 0.7 V for silicon and 0.3 V for germanium.

So although the diode is forward biased, only a very small current flows until the voltage rises past the knee voltage.

18.4 Why charge carriers are not present in the depletion region?

When the p-n junction forms, majority carriers diffuse across the junction: electrons from the n-side into the p-side and holes from the p-side into the n-side.

Near the junction they meet and recombine, so the mobile carriers disappear from that zone.

What is left behind are the immobile donor ions on the n-side and acceptor ions on the p-side.

This zone therefore has no free charge carriers and is called the depletion region.

18.5 What is the effect of forward and reverse biasing of a diode on the width of depletion region?

Forward biasing narrows the depletion region.

The applied voltage pushes majority carriers towards the junction, where they neutralize some of the exposed ions, so the depletion layer shrinks and current can flow.

Reverse biasing widens the depletion region.

The applied voltage pulls majority carriers away from the junction, uncovering more ions, so the depletion layer grows wider and almost no current flows.

18.6 Why ordinary silicon diodes do not emit light?

Light emission needs electron-hole recombination to release energy as photons.

Silicon has an indirect band gap, so when electrons and holes recombine in silicon, the energy is released mainly as heat (lattice vibrations) rather than as light.

Ordinary silicon diodes are also not made of transparent material and are not designed to let light out.

Light emitting diodes therefore use direct band-gap materials such as gallium arsenide, in which recombination gives photons.

18.7 Why a photo diode is operated in reverse biased state?

A photodiode is operated in reverse bias because its reverse current is very sensitive to light.

In reverse bias the normal current is tiny, so the extra current produced when light creates electron-hole pairs in the depletion region stands out clearly.

The photocurrent then grows almost linearly with light intensity, which makes the device a good light detector.

In forward bias the large forward current would completely swamp the small photocurrent.

Reverse bias also widens the depletion region, giving light a larger volume in which to create carriers.

18.8 Why is the base current in a transistor very small?

The base of a transistor is made very thin and very lightly doped.

When electrons are injected from the emitter into the base, almost all of them diffuse straight across the thin base into the collector.

Only a very small fraction, about one or two percent, recombine with holes in the base.

The base current is just this small recombination current, so it is very small compared with the collector and emitter currents.

18.9 What is the biasing requirement of the junctions of a transistor for its normal operation? Explain how these requirements are met in a common emitter amplifier?

For normal (active) operation, the emitter-base junction must be forward biased and the collector-base junction must be reverse biased.

In a common emitter amplifier the input signal is applied between base and emitter, with a base battery or bias network making the base-emitter junction forward biased.

The collector supply is connected so that the collector-base junction stays reverse

biased, with the collector at a higher potential than the base.

The amplified output is taken from the collector circuit, across the load resistor.

18.10 What is the principle of virtual ground? Apply it to find the gain of an inverting amplifier.

Virtual ground is the principle that, in an op-amp with negative feedback and very high open-loop gain, the inverting and non-inverting inputs sit at virtually the same potential.

If the non-inverting input is earthed (0 V), the inverting input is also at about 0 V, a virtual ground, although it is not actually connected to earth.

Apply it to the inverting amplifier: input voltage Vi drives current Ii = Vi / R1 into the inverting input node.

Almost no current enters the op-amp itself, so this current must flow through the feedback resistor Rf.

The output voltage is then Vo = -Ii x Rf = -(Vi / R1) x Rf.

So the gain is Vo / Vi = -Rf / R1. The minus sign shows the output is inverted.

18.11 The inputs of a gate are 1 and 0. Identify the gate if its output is (a) 0, (b) 1

(a) Output 0: the gate is an AND gate, because 1 AND 0 = 0.

An AND gate gives 1 only when all its inputs are 1.

(b) Output 1: the gate is an OR gate, because 1 OR 0 = 1.

An OR gate gives 1 when any one of its inputs is 1.

Exercise - 18.12 Tick the correct answer (book pp. 172-173, verbatim)

18.12 (i) A diode characteristic curve is a plot between

A. current and time

B. voltage and time

✓ C. voltage and current

D. forward voltage and reverse voltage

18.12 (ii) The colour of light emitted by a LED depends on

A. its forward bias

B. its reverse bias

C. the amount of forward current

✓ D. the type of semi-conductor material used

18.12 (iii) In a half-wave rectifier the diode conducts during

A. both halves of the input cycle

✓ B. a portion of the positive half of the input cycle

C. a portion of the negative half of the input cycle

D. one half of the input cycle

18.12 (iv) In a bridge rectifier of Fig. Q. 18.1 when Vi is positive at point B with respect to point A, which diodes are ON.

✓ A. D2 and D4

B. D1 and D3

C. D2 and D3

D. D1 and D4

18.12 (v) The common emitter current amplification factor beta is given by

A. Ic / Ie

✓ B. Ic / Ib

C. Ie / Ib

D. Ib / Ie

18.12 (vi) Truth table of logic function

A. summarizes its output values

B. tabulates all its input conditions only

✓ C. display all its input/output possibilities

D. is not based on logic algebra

18.12 (vii) The output of a two inputs OR gate is 0 only when its

✓ A. both inputs are 0

B. either input is 1

C. both inputs are 1

D. either input is 0

18.12 (viii) A two inputs NAND gate with inputs A and B has an output 0 if

A. A is 0

B. B is 0

C. both A and B are zero

✓ D. both A and B are 1

18.12 (ix) The truth table shown below is for (A B X: 0 0 1; 0 1 0; 1 0 0; 1 1 1)

✓ A. XNOR gate

B. OR gate

C. AND gate

D. NAND gate

Exercise - PROBLEMS 18.1 to 18.5 (book p. 174, verbatim, with the book's printed answers)

Problem 18.1 - The current flowing into the base of a transistor is 100 microA. Find its collector current Ic, its emitter current Ie and the ratio Ic / Ie, if the value of current gain beta is 100. (Book answers: 10 mA, 10.1 mA, 0.99)

Collector current: Ic = beta x Ib.

Ic = 100 x (100 x 10^-6) = 100 x 10^-4 = 10 x 10^-3 A = 10 mA.

Emitter current: Ie = Ib + Ic = 0.1 mA + 10 mA = 10.1 mA.

Ratio: Ic / Ie = 10 / 10.1 = 0.99.

Answers: 10 mA, 10.1 mA and 0.99, as printed in the book.

Problem 18.2 - Fig. P. 18.2 shows a transistor which operates a relay as the switch S is closed. The relay is energized by a current of 10 mA. Calculate the value Rb which will just make the relay operate. The current gain beta of the transistor is 200. When the transistor conducts, its Vbe can be assumed to be 0.6 V. (Fig: 9 V supply, base resistor Rb, switch S.) (Book answer: 168 k ohm)

Step 1: The relay needs a collector current Ic = 10 mA.

Step 2: Base current needed: Ib = Ic / beta = 10 mA / 200 = 0.05 mA = 50 microA.

Step 3: Voltage across the base resistor: V = supply voltage - Vbe = 9 - 0.6 = 8.4 V.

Step 4: Rb = V / Ib = 8.4 / (50 x 10^-6) = 168000 ohm.

Answer: 168 k ohm, as printed in the book.

Problem 18.3 - In circuit (Fig. P. 18.3), there is negligible potential drop between B and E, if beta is 100. Calculate (i) base current (ii) collector current (iii) potential drop across Rc (iv) Vce. (Fig: Vcc = 9 V, Rb = 800 k ohm, Rc = 1 k ohm.) (Book answers: 11.25 microA, 1.125 mA, 1.125 V, 7.875 V)

Step 1: (i) Base current: the base-emitter drop is negligible, so the full 9 V appears across Rb.

Ib = Vcc / Rb = 9 / (800 x 10^3) = 11.25 x 10^-6 A = 11.25 microA.

Step 2: (ii) Collector current: Ic = beta x Ib = 100 x 11.25 microA = 1.125 mA.

Step 3: (iii) Potential drop across Rc: V = Ic x Rc = (1.125 x 10^-3) x (1 x 10^3) = 1.125 V.

Step 4: (iv) Vce = Vcc - drop across Rc = 9 - 1.125 = 7.875 V.

Answers: 11.25 microA, 1.125 mA, 1.125 V and 7.875 V, as printed in the book.

Problem 18.4 - Calculate the output of the op-amp circuit shown in Fig. P. 18.4. (Fig: summing inverting amplifier, inputs +5 V through 10 k ohm and -2 V through 4 k ohm, feedback resistor 20 k ohm.) (Book answer: 0)

For an inverting summing amplifier: Vo = -Rf x (V1 / R1 + V2 / R2).

Here Rf = 20 k ohm, V1 = +5 V through R1 = 10 k ohm, V2 = -2 V through R2 = 4 k ohm.

V1 / R1 = 5 / (10 x 10^3) = 0.5 x 10^-3 A.

V2 / R2 = -2 / (4 x 10^3) = -0.5 x 10^-3 A.

Sum = 0.5 x 10^-3 - 0.5 x 10^-3 = 0.

Vo = -(20 x 10^3) x 0 = 0.

Answer: 0, as printed in the book.

Problem 18.5 - Calculate the gain of non-inverting amplifier shown in Fig. P. 18.5. (Fig: R1 = 10 k ohm to ground, feedback resistor Rf = 40 k ohm.) (Book answer: 5)

Gain of a non-inverting amplifier: A = 1 + Rf / R1.

Here Rf = 40 k ohm and R1 = 10 k ohm.

A = 1 + 40 / 10 = 1 + 4 = 5.

Answer: 5, as printed in the book.

Chapter 19: Dawn of Modern Physics

Exercise - QUESTIONS 19.1 to 19.27 (book pp. 199-200, verbatim)

19.1 What are the measurements on which two observers in relative motion will always agree upon?

Two observers in relative motion will always agree on the speed of light in free space, c.

This is the second postulate of special relativity: c has the same value for all observers, whatever their state of motion.

They will also agree on the form of the laws of physics, the first postulate.

But they will disagree on measurements of length, time intervals and simultaneity, which depend on their relative motion.

19.2 Does the dilation means that time really passes more slowly in moving system or that it only seems to pass more slowly?

Time really passes more slowly in the moving system; time dilation is a real physical effect, not an illusion.

It has been confirmed by experiment, for example fast-moving muons live much

longer than muons at rest, and atomic clocks flown around the Earth tick more slowly.

The effect is symmetric: each observer measures the other observer's clock as running slow.

There is no contradiction, because the two observers do not share the same notion of simultaneity.

19.3 If you are moving in a spaceship at a very high speed relative to the Earth, would you notice a difference (a) in your pulse rate (b) in the pulse rate of people on Earth?

(a) No, you would notice no difference in your own pulse rate.

In your own frame of reference you are at rest, and all your biological processes run normally; your pulse is your proper time.

(b) The pulse rate of people on Earth would appear slower to you.

Their clocks run slow as measured in your frame, by time dilation, so you would see their hearts beating more slowly than normal.

19.4 If the speed of light were infinite, what would the equations of special theory of relativity reduce to?

If c were infinite, the Lorentz factor gamma = 1 / sqrt(1 - v squared / c squared) would become 1 for any speed v.

Time dilation and length contraction would vanish, and moving clocks and rods would agree with stationary ones.

The Lorentz transformations would reduce to the Galilean transformations of classical mechanics.

Simultaneity would become absolute again, and Newtonian mechanics would be exactly correct at all speeds.

19.5 Since mass is a form of energy, can we conclude that a compressed spring has more mass than the same spring when it is not compressed?

Yes, in principle a compressed spring has more mass than the same spring uncompressed.

Compressing the spring stores elastic potential energy in it.

By E = m c squared, this extra energy corresponds to extra mass, delta m = (extra energy) / c squared.

The increase is far too small to measure, because c squared is enormous, but it is real: the mass of a body includes the mass equivalent of all its internal energy.

19.6 As a solid is heated and begins to glow, why does it first appear red?

As a solid is heated, the peak of its emitted radiation shifts to shorter wavelengths (Wien's displacement law).

At first the peak lies in the infrared, which we cannot see, and only the long-wavelength tail reaches into the visible.

The longest visible wavelength is red, so the first visible glow is red.

As the temperature rises further, the peak moves through orange and yellow towards white, as shorter wavelengths become strong.

19.7 What happens to total radiation from a blackbody if its absolute temperature is doubled?

By the Stefan-Boltzmann law, the total radiation emitted by a blackbody is proportional to the fourth power of its absolute temperature.

If the temperature is doubled, the total radiation increases by a factor of 2 to the power 4.

So the total radiation becomes 16 times greater.

19.8 A beam of red light and a beam of blue light have exactly the same energy. Which beam contains the greater number of photons?

The red beam contains the greater number of photons.

The energy of one photon is E = h f, and red light has a lower frequency than blue light.

So each red photon carries less energy than each blue photon.

To make up the same total energy, the red beam must contain more photons than the blue beam.

19.9 Which photon, red, green, or blue carries the most (a) energy and (b) momentum?

Blue light has the highest frequency of the three, then green, then red.

(a) Energy: a photon's energy is E = h f, so the blue photon carries the most energy.

(b) Momentum: a photon's momentum is p = E / c = h f / c, which is also greatest for the highest frequency.

So the blue photon carries the most momentum as well.

19.10 Which has the lower energy quanta? Radiowaves or X-rays

Radiowaves have the lower energy quanta.

A quantum of radiation has energy E = h f, so lower frequency means lower energy per quantum.

Radiowaves have far lower frequencies than X-rays.

So a radiowave photon carries far less energy than an X-ray photon.

19.11 Does the brightness of a beam of light primarily depends on the frequency of photons or on the number of photons?

The brightness of a beam depends primarily on the number of photons, that is, on the intensity.

More photons arriving per second means a brighter beam, whatever their individual energy.

The frequency of the photons decides their colour and the energy of each photon, not the brightness.

So brightness is governed by photon number, not by frequency.

19.12 When ultraviolet light falls on certain dyes, visible light is emitted. Why does this not happen when infrared light falls on these dyes?

This is fluorescence: the dye molecules absorb a photon, jump to an excited state, and then fall back, emitting visible light.

Ultraviolet photons carry enough energy to lift the molecules to the excited state.

Infrared photons carry too little energy to excite the molecules at all.

So with infrared light nothing is excited, and no visible light is emitted.

19.13 Will bright light eject more electrons from a metal surface than dimmer light of the same colour?

Yes, bright light ejects more electrons than dim light of the same colour.

Bright light contains more photons per second, and each photon can eject at most one electron.

So the number of photoelectrons per second, the photocurrent, is proportional to the intensity.

The colour (frequency) decides the maximum energy of each ejected electron, not how many are ejected.

19.14 Will higher frequency light eject greater number of electrons than low frequency light?

No, higher frequency does not by itself eject a greater number of electrons.

The number of electrons ejected per second depends on the number of photons per second, that is, on the intensity, not on the frequency.

Higher frequency raises the maximum kinetic energy of each ejected electron, by Einstein's photoelectric equation.

But if the intensity is the same, the count of electrons is the same, provided the frequency is above the threshold frequency.

19.15 When light shines on a surface, is momentum transferred to the metal surface?

Yes, momentum is transferred to the surface.

Photons carry momentum p = E / c = h / lambda, even though they have no rest

mass.

When photons strike the surface and are absorbed or reflected, they hand over some or all of this momentum.

This transfer of momentum is what produces radiation pressure on the surface.

19.16 Why can red light be used in a photographic dark room when developing films, but a blue or white light cannot?

Photographic film is sensitive to blue and white light but almost insensitive to red light.

Red photons have too little energy to trigger the chemical change in the silver halide grains of the film.

So a red safelight lets the worker see without fogging or exposing the film.

Blue or white light contains higher-energy photons that would expose the film and spoil it.

19.17 Photon A has twice the energy of photon B. What is the ratio of the momentum of A to that of B?

A photon's momentum is p = E / c, so momentum is directly proportional to energy.

If photon A has twice the energy of photon B, it has twice the momentum.

So the ratio of the momentum of A to that of B is 2 : 1.

19.18 Why don't we observe a Compton effect with visible light?

The Compton shift in wavelength is at most h / (m c) = 2.43 x 10^-12 m, a fixed amount independent of the incident wavelength.

For visible light, with wavelengths around 500 nm = 5 x 10^-7 m, this shift is about a hundred-thousandth of the wavelength.

The fractional change is far too small to detect.

With X-rays, whose wavelengths are comparable to the shift, the effect is easily observed.

19.19 Can pair production take place in vacuum? Explain

No, pair production cannot take place in empty vacuum.

A photon turning into an electron-positron pair must conserve both energy and momentum.

A single photon in vacuum cannot do both at once: the pair's total momentum cannot match the photon's momentum at the required total energy.

A nearby nucleus (or another particle) must be present to take up the recoil momentum.

So pair production happens near a nucleus, and the photon must have at least 1.02 MeV of energy.

19.20 Is it possible to create a single electron from energy? Explain.

No, a single electron cannot be created alone from energy.

Charge must be conserved: creating a negatively charged electron must also create an equal positive charge.

Energy therefore creates an electron-positron pair together, never a lone electron.

Lepton number is also conserved, and the pair has total lepton number zero, like the photon that made it.

19.21 If electrons behaved only like particles, what pattern would you expect on the screen after the electrons passes through the double slit?

If electrons behaved only like particles, there would be no interference pattern.

Each electron would pass through one slit or the other and land on the screen behind that slit.

The screen would show just two bright bands, two piles of electrons, one behind each slit.

The interference fringes actually observed prove that electrons also behave like waves.

19.22 If an electron and a proton have the same de Broglie wavelength, which particle has greater speed?

The electron has the greater speed.

De Broglie wavelength is lambda = h / p, so equal wavelengths mean equal momenta.

Momentum is p = m v, so the lighter particle must move faster to have the same momentum.

The electron is far lighter than the proton, so it has the greater speed.

19.23 We do not notice the de Broglie wavelength for a pitched cricket ball. Explain why?

The de Broglie wavelength of a cricket ball is unimaginably small.

Wavelength is lambda = h / (m v), and the ball's mass and speed make its momentum enormous compared with Planck's constant.

For a 0.16 kg ball at 40 m s^-1, lambda is of the order of 10^-34 m.

That is far smaller than any opening or obstacle, so no diffraction or interference can ever be observed; the ball behaves purely as a particle.

19.24 If the following particles have the same energy, which has the shortest wavelength? Electron, alpha particle, neutron, proton.

The alpha particle has the shortest wavelength.

For a particle of kinetic energy E, momentum is p = sqrt(2 m E), so wavelength lambda = h / sqrt(2 m E).

At the same energy, the heaviest particle has the largest momentum and hence the shortest wavelength.

The alpha particle (mass about 4 atomic mass units) is heavier than the proton and neutron (about 1 unit each) and far heavier than the electron.

So the alpha particle's wavelength is the shortest.

19.25 When does light behave as a wave? When does it behave as a particle?

Light behaves as a wave when it propagates and when it shows interference, diffraction and polarization.

These are explained by the wave theory, with light as an electromagnetic wave.

Light behaves as a particle when it interacts with matter and exchanges energy and momentum in discrete packets, photons.

The photoelectric effect, the Compton effect, and pair production are explained by photons, not by waves.

So light has a dual nature: wave-like in propagation, particle-like in interaction.

19.26 What advantages an electron microscope has over an optical microscope?

An electron microscope has a far higher resolving power than an optical microscope.

Resolution is limited by the wavelength used: an optical microscope uses visible light of wavelength about 500 nm and cannot resolve details much smaller than that.

An electron microscope uses electrons whose de Broglie wavelength is thousands of times shorter.

It can therefore resolve details down to atomic dimensions, far beyond any optical microscope.

19.27 If measurements show a precise position for an electron, can those measurements show precise momentum also? Explain.

No, a precise position measurement cannot also give a precise momentum.

Heisenberg's uncertainty principle says delta x x delta p is at least of the order of h / (2 pi): the more precisely position is known, the less precisely momentum can be known.

Pinning down the electron's position disturbs it and spreads out its momentum.

So position and momentum cannot both be measured with perfect accuracy at the same time.

Exercise - PROBLEMS 19.1 to 19.10 (book p. 201, verbatim, with the book's printed answers)

Problem 19.1 - A particle called the pion lives on the average only about 2.6 x 10^-8 s when at rest in the laboratory. It then changes to another form. How long would such a particle live when shooting through the space at 0.95 c? (Book answer: 8.3 x 10^-8 s)

Time dilation: moving lifetime = gamma x rest lifetime, with gamma = 1 / sqrt(1 - v squared / c squared).

Here v / c = 0.95, so gamma = 1 / sqrt(1 - 0.9025) = 1 / sqrt(0.0975) = 1 / 0.3122 = 3.20.

Moving lifetime = 3.20 x (2.6 x 10^-8) = 8.3 x 10^-8 s.

Answer: 8.3 x 10^-8 s, as printed in the book.

Problem 19.2 - What is the mass of a 70 kg man in a space rocket travelling at 0.8 c from us as measured from Earth? (Book answer: 116.7 kg)

Relativistic mass: m = m-zero / sqrt(1 - v squared / c squared).

Here v / c = 0.8, so sqrt(1 - 0.64) = sqrt(0.36) = 0.60.

m = 70 / 0.60 = 116.7 kg.

Answer: 116.7 kg, as printed in the book.

Problem 19.3 - Find the energy of photon in (b) Radiowave of wavelength 100 m (c) Green light of wavelength 550 nm (d) X-ray with wavelength 0.2 nm (Book answers: (a) 1.24 x 10^-6 eV (b) 2.25 eV (c) 6200 eV)

Photon energy: E = h c / lambda. With h c = 1240 eV nm, E (in eV) = 1240 / lambda (in nm).

Radiowave: the book's answer 1.24 x 10^-6 eV requires lambda = 1 m = 10^9 nm: E = 1240 / 10^9 = 1.24 x 10^-6 eV. (If the wavelength were literally 100 m, E would be 1.24 x 10^-8 eV; the stem's '100 m' is most plausibly the printed '10^0 m' = 1 m.)

Green light: lambda = 550 nm, E = 1240 / 550 = 2.25 eV.

X-ray: lambda = 0.2 nm, E = 1240 / 0.2 = 6200 eV.

Answers: 1.24 x 10^-6 eV, 2.25 eV and 6200 eV, as printed in the book (note the radiowave part above).

Problem 19.4 - Yellow light of 577 nm wavelength is incident on a cesium surface. The stopping voltage is found to be 0.25 V. Find (a) the Maximum K.E. of the photoelectrons (b) the work function of cesium (Book answers: (a) 4 x 10^-20 J (b) 1.91 eV)

Step 1: (a) Maximum kinetic energy = e x stopping voltage = 0.25 eV.

In joules: 0.25 x (1.60 x 10^-19) = 4.0 x 10^-20 J.

Step 2: Photon energy: E = h c / lambda = 1240 / 577 = 2.15 eV.

Step 3: (b) Work function: phi = E - K.E.-max = 2.15 - 0.25 = 1.90 eV by this rounding; the book prints 1.91 eV.

Answers: (a) 4 x 10^-20 J and (b) 1.91 eV, as printed in the book.

Problem 19.5 - X-rays of wavelength 22 pm are scattered from a carbon target. The scattered radiation being viewed at 85 degrees to the incident beam. What is Compton shift? (Book answer: 2.2 x 10^-12 m)

Compton shift: change in lambda = (h / (m c)) x (1 - cos theta), with h / (m c) = 2.43 x 10^-12 m.

Here theta = 85 degrees, cos 85 degrees = 0.0872.

Change in lambda = (2.43 x 10^-12) x (1 - 0.0872) = (2.43 x 10^-12) x 0.9128.

Change in lambda = 2.2 x 10^-12 m.

Answer: 2.2 x 10^-12 m, as printed in the book.

Problem 19.6 - A 90 keV X-ray photon is fired at a carbon target and Compton scattering occurs. Find the wavelength of the incident photon and the wavelength of the scattered photon for scattering angle of (a) 30 degrees (b) 60 degrees (Book answers: 13.8 pm (a) 14.1 pm (b) 15 pm)

Step 1: Incident wavelength: lambda = h c / E = 1240 eV nm / 90000 eV = 0.0138 nm = 13.8 pm.

Step 2: Compton shift: change in lambda = (2.43 pm) x (1 - cos theta).

Step 3: (a) theta = 30 degrees: change = 2.43 x (1 - 0.866) = 2.43 x 0.134 = 0.33 pm.

Scattered wavelength = 13.8 + 0.33 = 14.1 pm.

Step 4: (b) theta = 60 degrees: change = 2.43 x (1 - 0.500) = 2.43 x 0.500 = 1.22 pm.

Scattered wavelength = 13.8 + 1.22 = 15.0 pm.

Answers: 13.8 pm, (a) 14.1 pm and (b) 15 pm, as printed in the book.

Problem 19.7 - What is the maximum wavelength of the two photons produced when a positron annihilates an electron? The rest mass energy of each is 0.51 MeV. (Book answer: 2.44 x 10^-12 m)

In annihilation the total rest energy, 2 x 0.51 = 1.02 MeV, appears as two photons.

The longest wavelength corresponds to the smallest photon energy, when the energy is shared equally: each photon gets 0.51 MeV.

Wavelength: lambda = h c / E = 1240 MeV fm / 0.51 MeV = 2431 fm.

Answer: 2.44 x 10^-12 m, as printed in the book.

Problem 19.8 - Calculate the wavelength of (a) a 140 g ball moving at 40 m s^-1 (b) a proton moving at the same speed (c) an electron moving at the same speed (Book answers: (a) 1.18 x 10^-34 m (b) 9.92 nm (c) 1.82 x 10^-5 m)

De Broglie wavelength: lambda = h / (m v), with h = 6.63 x 10^-34 J s and v = 40 m s^-1.

(a) Ball: m = 140 g = 0.14 kg. lambda = (6.63 x 10^-34) / (0.14 x 40) = (6.63 x 10^-34) / 5.6 = 1.18 x 10^-34 m.

(b) Proton: m = 1.67 x 10^-27 kg. lambda = (6.63 x 10^-34) / ((1.67 x 10^-27) x 40) = (6.63 x 10^-34) / (6.68 x 10^-26) = 9.92 x 10^-9 m = 9.92 nm.

(c) Electron: m = 9.11 x 10^-31 kg. lambda = (6.63 x 10^-34) / ((9.11 x 10^-31) x 40) = (6.63 x 10^-34) / (3.64 x 10^-29) = 1.82 x 10^-5 m.

Answers: (a) 1.18 x 10^-34 m, (b) 9.92 nm and (c) 1.82 x 10^-5 m, as printed in the book.

Problem 19.9 - What is the de Broglie wavelength of an electron whose kinetic energy is 120 eV? (Book answer: 1.12 x 10^-10 m)

Momentum from kinetic energy: p = sqrt(2 m K).

K = 120 eV = 120 x (1.60 x 10^-19) = 1.92 x 10^-17 J. m = 9.11 x 10^-31 kg.

p = sqrt(2 x (9.11 x 10^-31) x (1.92 x 10^-17)) = sqrt(3.50 x 10^-47) = 5.92 x 10^-24 kg m s^-1.

Wavelength: lambda = h / p = (6.63 x 10^-34) / (5.92 x 10^-24) = 1.12 x 10^-10 m.

Answer: 1.12 x 10^-10 m, as printed in the book.

Problem 19.10 - An electron is placed in a box about the size of an atom that is

about 1.0 x 10^-10 m. What is the velocity of the electron? (Book answer: 7.29 x 10^6 m s^-1)

Uncertainty principle (book's form): delta x x delta p is of the order of h.

With delta x = 1.0 x 10^-10 m: delta p = h / delta x = (6.63 x 10^-34) / (1.0 x 10^-10) = 6.63 x 10^-24 kg m s^-1.

Velocity: v = delta p / m = (6.63 x 10^-24) / (9.11 x 10^-31) = 7.28 x 10^6 m s^-1.

Answer: 7.29 x 10^6 m s^-1, as printed in the book.

Chapter 20: Atomic Spectra

Exercise - QUESTIONS 20.1 to 20.10 (book p. 217, verbatim)

20.1 Bohr's theory of hydrogen atom is based upon several assumptions. Do any of these assumptions contradict classical physics?

Yes, two of Bohr's assumptions contradict classical physics.

First, Bohr assumed that an electron in a stationary orbit does not radiate energy, although classical electrodynamics says an accelerating (circularly moving) charge must radiate and spiral into the nucleus.

Second, he assumed that the electron's angular momentum is quantized, L = n h / (2 pi), whereas classical physics allows angular momentum to take any value.

Bohr's theory was therefore a hybrid: classical mechanics for the orbits, plus quantum rules imposed on them.

20.2 What is meant by a line spectrum? Explain, how line spectrum can be used for

the identification of elements?

A line spectrum is a spectrum consisting of separate bright lines at definite wavelengths, with darkness in between, instead of a continuous rainbow.

It is produced when excited atoms emit light: each line corresponds to a transition between two energy levels of the atom.

Every element has its own unique set of energy levels, so its line pattern is unique, like a fingerprint.

By matching an unknown sample's lines with known spectra, the elements present in the sample can be identified.

20.3 Can the electron in the ground state of hydrogen absorb a photon of energy 13.6 eV and greater than 13.6 eV?

Yes, it can absorb both.

The ground-state energy of hydrogen is -13.6 eV, so a photon of exactly 13.6 eV gives the electron just enough energy to escape the atom: the atom is ionized.

A photon of energy greater than 13.6 eV is also absorbed, and the extra energy beyond 13.6 eV becomes the kinetic energy of the freed electron.

A photon of less than 13.6 eV, not equal to any level difference, cannot be absorbed at all.

20.4 How can the spectrum of hydrogen contain so many lines when hydrogen contains one electron?

Each hydrogen atom has only one electron, but a glowing sample contains a huge number of atoms.

At any moment, different atoms have their electrons in different excited states.

Each atom's electron can drop through many possible pairs of levels, and each possible transition gives one spectral line.

The spectrum we see is the combined light of all the atoms, so it contains lines for every allowed transition, even though each atom contributes only one line at a

time.

20.5 Is energy conserved when an atom emits a photon of light?

Yes, energy is fully conserved when an atom emits a photon.

The photon's energy is exactly the difference between the atom's higher and lower energy levels: h f = E-higher - E-lower.

The atom's internal energy falls by exactly the amount the photon carries away.

Total energy, atom plus photon, stays the same; the tiny recoil energy of the atom is negligible.

20.6 Explain why a glowing gas gives only certain wavelengths of light and why that gas is capable of absorbing the same wavelengths? Give a reason why it is transparent to other wavelengths?

A glowing gas emits light when its excited electrons fall to lower energy levels.

Each fall releases a photon whose energy equals the difference of the two levels, so only the wavelengths matching level differences appear: the emission spectrum is a set of lines.

The same gas absorbs light when a ground-state electron jumps up by absorbing a photon of exactly a level-difference energy.

So it absorbs the same wavelengths it emits.

Other wavelengths match no level difference, so no electron can absorb them; they pass straight through, and the gas is transparent to them.

20.7 What do we mean when we say that the atom is excited?

An atom is excited when one of its electrons has been raised from the ground state to a higher energy level.

This happens when the atom absorbs energy, for example from a photon or a collision.

The excited state is unstable, and after a short time the electron falls back, emitting the extra energy as a photon.

20.8 Can X-rays be reflected, refracted, diffracted and polarized just like any other waves? Explain.

Yes, X-rays show all these wave properties, because they are electromagnetic waves like visible light.

They can be reflected and refracted, though the effects are small because their wavelengths are so short.

They are diffracted by crystal lattices, whose atomic spacings are comparable to X-ray wavelengths; this is the basis of X-ray crystallography.

They can also be polarized, for example by scattering, just like visible light.

20.9 What are the advantages of lasers over ordinary light?

Lasers are monochromatic: they emit light of essentially a single wavelength, unlike ordinary light which mixes many wavelengths.

Lasers are coherent: all their waves are in phase with each other, so they interfere constructively and stay in step.

Lasers are highly directional: the beam spreads very little and stays narrow over long distances.

Lasers are very intense: a large power can be concentrated into a tiny area, which is why they can cut, drill and carry signals.

20.10 Explain why laser action could not occur without population inversion between atomic levels?

Laser action needs stimulated emission to outnumber absorption.

Normally most atoms sit in the lower level, so an incoming photon is more likely to be absorbed by a lower-level atom than to stimulate an upper-level atom.

Population inversion reverses this: more atoms are in the upper level than in the lower level.

Then stimulated emission dominates, each photon produces two identical photons, and the beam is amplified.

Without inversion there is net absorption, not amplification, and no laser action.

Exercise - PROBLEMS 20.1 to 20.10 (book pp. 217-218, verbatim, with the book's printed answers)

Problem 20.1 - A hydrogen atoms is in its ground state (n = 1). Using Bohr's theory, calculate (a) the radius of the orbit, (b) the linear momentum of the electron, (c) the angular momentum of the electron (d) the kinetic energy (e) the potential energy, and (f) the total energy. (Book answers: (a) 0.529 x 10^-10 m (b) 1.99 x 10^-24 kg m s^-1 (c) 1.05 x 10^-34 kg m squared s^-1 (d) 13.6 eV (e) -27.2 eV (f) -13.6 eV)

Step 1: (a) Bohr radius for n = 1: r1 = 0.529 x 10^-10 m.

Step 2: (b) Speed in the ground state: v = 2.18 x 10^6 m s^-1. Linear momentum p = m v = (9.11 x 10^-31) x (2.18 x 10^6).

p = 1.99 x 10^-24 kg m s^-1.

Step 3: (c) Angular momentum: L = n h / (2 pi) = (6.63 x 10^-34) / (2 pi) = 1.05 x 10^-34 kg m squared s^-1.

Step 4: (d) Kinetic energy = +13.6 eV (the electron's motion energy in the ground state).

Step 5: (e) Potential energy = -2 x kinetic energy = -27.2 eV.

Step 6: (f) Total energy = kinetic + potential = 13.6 - 27.2 = -13.6 eV.

Answers: (a) 0.529 x 10^-10 m, (b) 1.99 x 10^-24 kg m s^-1, (c) 1.05 x 10^-34 kg

m squared s^-1, (d) 13.6 eV, (e) -27.2 eV and (f) -13.6 eV, as printed in the book.

Problem 20.2 - What are the energies in eV of quanta of wavelength? lambda = 400, 500 and 700 nm. (Book answers: 3.10 eV, 2.49 eV, 1.77 eV)

Photon energy: E = h c / lambda, with h c = 1240 eV nm.

For 400 nm: E = 1240 / 400 = 3.10 eV.

For 500 nm: E = 1240 / 500 = 2.48 eV by this value of h c; the book prints 2.49 eV.

For 700 nm: E = 1240 / 700 = 1.77 eV.

Answers: 3.10 eV, 2.49 eV and 1.77 eV, as printed in the book.

Problem 20.3 - An electron jumps from a level Ei = -3.5 x 10^-19 J to Ef = -1.20 x 10^-18 J. What is the wavelength of the emitted light? (Book answer: 234 nm)

The electron falls from the higher level Ei = -3.5 x 10^-19 J to the lower level Ef = -1.20 x 10^-18 J.

Photon energy = Ei - Ef = (-3.5 x 10^-19) - (-1.20 x 10^-18) = 8.5 x 10^-19 J.

Wavelength: lambda = h c / E = (6.63 x 10^-34 x 3.0 x 10^8) / (8.5 x 10^-19).

lambda = (1.989 x 10^-25) / (8.5 x 10^-19) = 2.34 x 10^-7 m.

Answer: 234 nm, as printed in the book.

Problem 20.4 - Find the wavelength of the spectral line corresponding to the transition in hydrogen from n = 6 state to n = 3 state? (Book answer: 1094 nm)

Rydberg formula: 1 / lambda = R (1 / nf squared - 1 / ni squared), with nf = 3, ni = 6, R = 1.097 x 10^7 m^-1.

1 / lambda = (1.097 x 10^7) x (1 / 9 - 1 / 36) = (1.097 x 10^7) x (0.1111 - 0.0278).

1 / lambda = (1.097 x 10^7) x 0.0833 = 9.14 x 10^5 m^-1.

lambda = 1 / (9.14 x 10^5) = 1.094 x 10^-6 m.

Answer: 1094 nm, as printed in the book.

Problem 20.5 - Compute the shortest wavelength radiation in the Balmer series? What value of n must be used? (Book answers: 364.5 nm, n = infinity)

Balmer series: nf = 2. The shortest wavelength comes from the largest energy jump, ni = infinity.

1 / lambda = R (1 / 4 - 1 / infinity) = R / 4 = (1.097 x 10^7) / 4 = 2.74 x 10^6 m^-1.

lambda = 1 / (2.74 x 10^6) = 3.645 x 10^-7 m.

Answers: 364.5 nm and n = infinity, as printed in the book.

Problem 20.6 - Calculate the longest wavelength of radiation for the Paschen series. (Book answer: 1875 nm)

Paschen series: nf = 3. The longest wavelength comes from the smallest energy

jump, ni = 4.

1 / lambda = R (1 / 9 - 1 / 16) = (1.097 x 10^7) x (0.1111 - 0.0625).

1 / lambda = (1.097 x 10^7) x 0.0486 = 5.33 x 10^5 m^-1.

lambda = 1 / (5.33 x 10^5) = 1.875 x 10^-6 m.

Answer: 1875 nm, as printed in the book.

Problem 20.7 - Electrons in an X-ray tube are accelerated through a potential difference of 3000 V. If these electrons were slowed down in a target, what will be the minimum wavelength of X-rays produced? (Book answer: 4.14 x 10^-10 m)

The most energetic X-ray photon takes the electron's full kinetic energy: E = e V = 3000 eV.

Minimum wavelength: lambda-min = h c / E = 1240 eV nm / 3000 eV = 0.413 nm.

lambda-min = 4.13 x 10^-10 m, printed by the book as 4.14 x 10^-10 m.

Answer: 4.14 x 10^-10 m, as printed in the book.

Problem 20.8 - The wavelength of K X-ray from copper is 1.377 x 10^-10 m. What is the energy difference between the two levels from which this transition results? (Book answer: 9.03 keV)

Energy difference: delta E = h c / lambda.

lambda = 1.377 x 10^-10 m = 0.1377 nm.

delta E = 1240 / 0.1377 = 9005 eV = 9.01 keV by this value of h c; the book prints 9.03 keV.

Answer: 9.03 keV, as printed in the book.

Problem 20.9 - A tungsten target is struck by electrons that have been accelerated from rest through 40 kV potential difference. Find the shortest wavelength of the bremsstrahlung radiation emitted. (Book answer: 0.31 x 10^-10 m)

The shortest wavelength photon takes the electron's full energy: E = 40 keV.

lambda-min = h c / E = 1240 eV nm / 40000 eV = 0.031 nm.

Answer: 0.31 x 10^-10 m, as printed in the book.

Problem 20.10 - The orbital electron of a hydrogen atom moves with a speed of 5.456 x 10^5 m s^-1. (a) Find the value of the quantum number n associated with this electron. (b) Calculate the radius of this orbit. (c) Find the energy of the electron in this orbit. (Book answers: n = 4, rn = 0.846 nm; En = -0.85 eV)

Step 1: (a) Bohr speed: vn = (2.18 x 10^6) / n, so n = (2.18 x 10^6) / (5.456 x 10^5) = 4.0.

Step 2: (b) Radius: rn = n squared x r1 = 16 x (0.529 x 10^-10) = 8.46 x 10^-10 m = 0.846 nm.

Step 3: (c) Energy: En = -13.6 / n squared = -13.6 / 16 = -0.85 eV.

Answers: n = 4, rn = 0.846 nm and En = -0.85 eV, as printed in the book.

Chapter 21: Nuclear Physics

Exercise - QUESTIONS 21.1 to 21.20 (book p. 258, verbatim)

21.1 What are isotopes? What do they have in common and what are their differences?

Isotopes are nuclei of the same element that have the same atomic number Z but different mass numbers A.

In common: the same number of protons, the same number of electrons in the neutral atom, the same chemical properties, and the same place in the periodic table.

Differences: different numbers of neutrons, different masses, and often different nuclear stability: some isotopes are stable while others are radioactive.

21.2 Why are heavy nuclei unstable?

In a heavy nucleus there are many protons, and the repulsive Coulomb force between them grows rapidly with proton number.

The attractive strong nuclear force is short-ranged and saturates: each nucleon only attracts its near neighbours.

In heavy nuclei the repulsion eventually overpowers the attraction, so the nucleus cannot hold together permanently.

Heavy nuclei therefore decay by emitting radiation until they reach a more stable configuration.

21.3 If a nucleus has a half-life of 1 year, does this mean that it will be completely decayed after 2 years? Explain.

No, it does not mean that.

Half-life is a statistical law for a large sample, not a lifetime for one nucleus.

After 1 year, half of the sample remains; after 2 years, which is two half-lives, one quarter of the original sample still remains and three quarters have decayed.

The sample never becomes exactly zero: each half-life only halves what is left, and any single nucleus may survive many half-lives.

21.4 What fraction of a radioactive sample decays after two half-lives have elapsed?

After one half-life, 1/2 of the sample remains.

After two half-lives, 1/2 of that half remains, which is 1/4 of the original sample.

So the fraction that has decayed is 1 - 1/4 = 3/4 of the sample.

21.5 The radioactive element 226_88 Ra has a half-life of 1.6 x 10^3 years. Since the Earth is about 5 billion years old, how can you explain why we still can find this element in nature?

Radium-226 cannot be primordial: with a half-life of only 1600 years, any radium present when the Earth formed would have decayed away completely long ago.

It is still found because it is continuously being made.

Radium-226 is a daughter product in the uranium-238 decay series: uranium-238, with a half-life of 4.5 billion years, decays step by step through thorium and other elements to radium-226.

Fresh radium-226 is therefore constantly produced in uranium ores, which is why it is still present in nature.

21.6 Describe a brief account of interaction of various types of radiations with matter.

Alpha radiation consists of heavy, doubly charged particles: they interact very strongly with matter, ionize densely along a short track, and are stopped by a sheet of paper or a few centimetres of air.

Beta radiation consists of fast electrons: they are much lighter and singly charged,

so they ionize less densely and travel further, being stopped by a few millimetres of aluminium.

Gamma radiation consists of uncharged photons: they do not ionize directly but knock electrons out of atoms by the photoelectric effect, Compton scattering and pair production, and those electrons then cause ionization.

Gamma rays are the most penetrating and need thick lead or concrete to stop them.

21.7 Explain how alpha and beta-particles may ionize an atom without directly hitting the electrons? What is the difference in the action of the two particles for producing ionization?

A charged particle need not strike an electron directly to ionize an atom.

Its electric field acts at a distance: as it passes near an atom, it pulls or pushes the atomic electrons with a Coulomb force.

If the energy transferred is enough, an electron is torn away and the atom is ionized.

The difference is in strength and density: an alpha particle carries charge +2e and moves slowly, so its field acts strongly and for longer, producing dense ionization along a short track.

A beta particle carries charge -e and moves much faster, so it produces sparser ionization spread over a longer track.

21.8 A particle which produces more ionization is less penetrating. Why?

Ionization costs the particle energy: every ion pair it creates takes energy from the particle.

A strongly ionizing particle loses its energy very quickly, so it is brought to rest after a short distance.

A weakly ionizing particle loses energy slowly and keeps most of it, so it travels much further into matter.

That is why alpha particles, which ionize most densely, penetrate least, while gamma rays, which ionize least directly, penetrate most.

21.9 What information is revealed by the length and shape of the tracks of an incident particle in Wilson cloud chamber?

The length of a track tells us the particle's range, and hence its energy: a longer track means a more energetic, more penetrating particle.

The thickness or density of droplets along the track tells us the ionizing power: a thick, dense track is an alpha particle, a thin, faint track is a beta particle.

The shape tells us about scattering: beta tracks are zigzag because the light electron is easily deflected, while alpha tracks are straight.

If a magnetic field is applied, the curvature of the track gives the sign of the charge and the momentum of the particle.

21.10 Why must a Geiger Muller tube for detecting alpha-particles have a very thin end window? Why does a Geiger Muller tube for detecting gamma-rays not need a window at all?

Alpha particles are stopped by the thinnest of materials, even a sheet of paper.

If the tube's end window were thick, the alpha particles would be absorbed in the window before reaching the gas inside, so the window must be extremely thin.

Gamma rays are so penetrating that they pass straight through the metal wall of the tube itself.

They are detected when they knock electrons out of the wall or gas inside the tube, so no window is needed at all.

21.11 Describe the principle of operation of a solid state detector of ionizing radiation in terms of generation and detection of charge carriers.

In a solid state detector, ionizing radiation enters a semiconductor crystal, such as silicon.

The radiation's energy creates electron-hole pairs in the crystal: each absorbed

chunk of energy frees charge carriers.

An applied voltage sweeps the electrons to one electrode and the holes to the other. The collected charge forms an electrical pulse whose size is proportional to the energy deposited by the radiation, so the detector both counts the particles and measures their energy.

21.12 What do we mean by the term critical mass?

Critical mass is the minimum mass of fissile material, such as uranium-235, needed to sustain a nuclear chain reaction.

In a smaller mass, too many neutrons escape through the surface, so on average fewer than one neutron from each fission causes another fission and the reaction dies out.

At the critical mass, on average at least one neutron per fission goes on to cause another fission, so the chain reaction sustains itself.

21.13 Discuss the advantages and disadvantages of nuclear power compared to the use of fossil fuel generated power.

Advantages of nuclear power: it produces no carbon dioxide or smoke during operation, so it does not add to global warming or air pollution.

A very small mass of fuel gives a very large amount of energy, so fuel transport and storage are easy.

Disadvantages: the waste stays radioactive for thousands of years and must be stored safely.

Accidents can release radioactivity, plants are expensive to build and to decommission, and the technology raises proliferation concerns.

Fossil fuel plants are cheaper and well established, but they burn huge quantities of fuel, emit carbon dioxide and pollutants, and depend on finite reserves.

21.14 What factors make a fusion reaction difficult to achieve?

Fusion is difficult because the nuclei to be fused both carry positive charge and

repel each other strongly.

To overcome this Coulomb repulsion they must be heated to tens of millions of degrees, so that they move fast enough to collide and fuse.

At such temperatures the fuel is a plasma that must be confined without touching any wall, which is done with magnetic fields or by inertial confinement.

It is also hard to get more energy out of the reaction than is put in to heat and confine the plasma, and the intense neutrons produced damage the reactor materials.

21.15 Discuss the advantages and disadvantages of fission power from the point of safety, pollution and resources.

Safety: a fission reactor is designed with control rods, shielding and containment, but an accident or loss of cooling can release radioactive material, so strict safety systems are essential.

Pollution: fission power emits no carbon dioxide or smoke, but it produces long-lived radioactive waste that must be isolated for thousands of years, and uranium mining also causes environmental damage.

Resources: uranium is a finite resource, but because fission releases so much energy per kilogram, a small amount of fuel lasts a long time, and breeder reactors can extend the supply further.

21.16 What do you understand by "background radiation"? State two sources of this radiation.

Background radiation is the low level of ionizing radiation that is always present around us, from natural and man-made sources.

Two sources are: cosmic rays arriving from outer space, and natural radioactivity in rocks and soil, including radon gas that seeps into buildings from the ground.

21.17 If someone accidently swallows an alpha-source and a beta-source which would be the more dangerous to him? Explain why?

The alpha source would be more dangerous.

Outside the body, alpha particles are harmless because they are stopped by the skin.

Inside the body, the alpha emitter sits in direct contact with living tissue and deposits all of its energy in a tiny volume, causing intense local damage: alpha radiation has a high relative biological effectiveness.

A beta source inside the body is also dangerous, but beta particles are less ionizing and spread their energy over a larger volume, so the local damage is less severe.

21.18 Which radiation dose would deposit more energy to the body (a) 10 mGy to the hand, or (b) 1 mGy dose to the entire body.

Energy deposited = absorbed dose x mass of the tissue.

(a) Hand: mass about 0.5 kg, so energy = (10 x 10^-3) x 0.5 = 5 x 10^-3 J.

(b) Whole body: mass about 70 kg, so energy = (1 x 10^-3) x 70 = 70 x 10^-3 J.

So the whole-body dose deposits more total energy, about fourteen times as much, even though the hand dose is locally more intense.

Answer: (b), the 1 mGy dose to the entire body.

21.19 What is a radioactive tracer? Describe one application each in medicine, agriculture and industry.

A radioactive tracer is a small amount of a radioactive isotope used to follow a process: its radiation can be detected from outside, showing where the substance goes.

Medicine: iodine-131 is used to diagnose and treat thyroid disorders, because iodine concentrates in the thyroid gland.

Agriculture: phosphorus-32 is used to study how plants take up fertilizer from the soil.

Industry: tracers are injected into pipelines to locate leaks, or used to measure wear in engines by the radioactivity appearing in the lubricating oil.

21.20 How can radioactivity help in the treatment of cancer?

Cancer cells divide rapidly and repair radiation damage poorly, so they are more sensitive to radiation than most normal cells.

In radiotherapy, a carefully aimed beam of gamma rays, for example from cobalt-60, or a radioactive isotope placed inside the body, such as iodine-131 for thyroid cancer, damages the DNA of the cancer cells.

The damaged cells die or stop dividing, while the surrounding healthy tissue is spared as much as possible by precise targeting and by splitting the treatment into small doses.

Exercise - PROBLEMS 21.1 to 21.10 (book p. 259, verbatim, with the book's printed answers; masses from the book's table on book p. 222)

Problem 21.1 - Find the mass defect and the binding energy for tritium, if the atomic mass of tritium is 3.016049 u. (Book answers: 0.00857 u, 7.97 MeV)

Tritium has 1 proton and 2 neutrons. From the book's table (book p. 222): proton mass = 1.007276 u, neutron mass = 1.008665 u.

Mass of the separate nucleons = 1.007276 + 2 x 1.008665 = 1.007276 + 2.017330 = 3.024606 u.

Mass defect = 3.024606 - 3.016049 = 0.008557 u, which the book rounds to 0.00857 u.

Binding energy = mass defect x 931.5 MeV/u = 0.00857 x 931.5 = 7.98 MeV, printed by the book as 7.97 MeV.

Answers: mass defect 0.00857 u and binding energy 7.97 MeV, as printed in the

book.

Problem 21.2 - The half-life of 91_38 Sr is 9.70 hours. Find its decay constant. (Book answer: 1.99 x 10^-5 s^-1)

Decay constant: lambda = 0.693 / T-half.

Half-life in seconds: 9.70 hours x 3600 = 34920 s.

lambda = 0.693 / 34920 = 1.98 x 10^-5 s^-1, printed by the book as 1.99 x 10^-5 s^-1.

Answer: 1.99 x 10^-5 s^-1, as printed in the book.

Problem 21.3 - The element 234_91 Pa is unstable and decays by beta-emission with a half-life 6.66 hours. State the nuclear reaction and the daughter nuclei. (Book answer: 234_92 U)

In beta-minus decay a neutron changes into a proton and an electron: the mass number stays the same and the atomic number increases by 1.

Reaction: 234_91 Pa -> 234_92 U + 0_-1 e.

The daughter nucleus is uranium-234: 234_92 U, as printed in the book.

Problem 21.4 - Find the energy associated with the following reaction: (Mass of 1H = 1.00784 u) 14_7 N + 4_2 He -> 17_8 O + 1_1 H. What does negative sign indicate? (Book answer: 1.12 MeV)

From the book's table (book p. 222): nitrogen-14 = 14.0031 u, helium-4 = 4.0026 u,

oxygen-17 = 16.9991 u; hydrogen-1 = 1.0078 u as given in the stem.

Mass of reactants = 14.0031 + 4.0026 = 18.0057 u.

Mass of products = 16.9991 + 1.0078 = 18.0069 u.

Mass change = 18.0057 - 18.0069 = -0.0012 u.

Energy: Q = (-0.0012) x 931.5 = -1.12 MeV.

The negative sign means the reaction is endoergic: the products are heavier than the reactants, so energy must be supplied, by the kinetic energy of the incoming alpha particle, for the reaction to happen.

Answer: -1.12 MeV (magnitude 1.12 MeV as printed in the book).

Problem 21.5 - Determine the energy associated with the following reaction: (mass of 14C = 14.0077 u) 14_6 C -> 14_7 N + 0_-1 e. (Book answer: 3.77 MeV)

Parent: carbon-14 = 14.0077 u, as given in the stem.

Daughter: nitrogen-14 = 14.0031 u from the book's table (book p. 222); the book's working takes the mass difference as 14.0077 - 14.0031 - 0.00055 (one electron mass) = 0.00405 u.

Energy = 0.00405 x 931.5 = 3.77 MeV.

Answer: 3.77 MeV, as printed in the book.

Note: the stem's carbon-14 mass of 14.0077 u is the book's own figure; with the modern atomic mass of carbon-14 (14.0032 u) this decay energy would be only about 0.156 MeV.

Problem 21.6 - If 233_92 U decays twice by alpha-emission, what is the resulting isotope? (Book answer printed as 225_88 Rn)

Each alpha emission reduces the mass number by 4 and the atomic number by 2.

After the first alpha: 233 - 4 = 229, 92 - 2 = 90, giving thorium-229.

After the second alpha: 229 - 4 = 225, 90 - 2 = 88.

The resulting isotope has mass number 225 and atomic number 88, which is radium-225: 225_88 Ra.

Note: the book prints the symbol as Rn, but atomic number 88 is radium (Ra); radon is atomic number 86. The numbers 225 and 88 are correct for radium-225.

Problem 21.7 - Calculate the energy (in MeV) released in the following fusion reaction: 2_1 H + 3_1 H -> 4_2 He + 1_0 n. (Book answer: 17.6 MeV/event)

From the book's table (book p. 222): deuterium = 2.014102 u, tritium = 3.01605 u, helium-4 = 4.002603 u, neutron = 1.008665 u.

Mass of reactants = 2.014102 + 3.01605 = 5.030152 u.

Mass of products = 4.002603 + 1.008665 = 5.011268 u.

Mass defect = 5.030152 - 5.011268 = 0.018884 u.

Energy released = 0.018884 x 931.5 = 17.59 MeV.

Answer: 17.6 MeV per event, as printed in the book.

Problem 21.8 - A sheet of lead 5.0 mm thick reduces the intensity of a beam of gamma-rays by a factor 0.4. Find half value thickness of lead sheet which will reduce the intensity to half of its initial value. (Book answer: 3.79 mm)

Intensity after thickness x: I = I0 e^(-mu x), so 0.4 = e^(-mu x 5.0).

Taking natural logs: ln(0.4) = -mu x 5.0, so mu = 0.9163 / 5.0 = 0.18326 mm^-1.

Half-value thickness: I / I0 = 1/2 = e^(-mu x-half), so x-half = ln 2 / mu = 0.693 / 0.18326 = 3.78 mm.

Answer: 3.79 mm, as printed in the book.

Problem 21.9 - Radiation from a point source obeys the inverse square law. If the count rate at a distance of 1.0 m from Geiger counter is 360 counts per minute, what will be its count rate at 3.0 m from the source? (Book answer: 40 counts per min.)

Inverse square law: count rate is proportional to 1 / (distance squared).

At 3.0 m the distance is 3 times larger, so the count rate falls by a factor of 3 squared = 9.

Count rate = 360 / 9 = 40 counts per minute.

Answer: 40 counts per minute, as printed in the book.

Problem 21.10 - A 75 kg person receives a whole body radiation dose of 24 m-rad, delivered by alpha-particles for which RBE factor is 12. Calculate (a) the absorbed energy in joules, and (b) the equivalent dose in rem. (Book answers: (a) 18 mJ (b) 0.29 rem)

Step 1: (a) Convert the absorbed dose: 24 m-rad = 0.024 rad. Since 1 rad = 0.01 Gy = 0.01 J/kg, the dose is 0.024 x 0.01 = 2.4 x 10^-4 J/kg.

Step 2: Absorbed energy = dose x mass = (2.4 x 10^-4) x 75 = 0.018 J = 18 mJ.

Step 3: (b) Equivalent dose = absorbed dose in rad x RBE = 0.024 x 12 = 0.288 rem.

Answers: (a) 18 mJ and (b) 0.29 rem, as printed in the book.